GENERATIVE AI FOR COMPUTER VISION AND MEDICAL IMAGE ANALYSIS: A SURVEY OF TASKS, TRENDS, AND RESEARCH OPPORTUNITIES

Fengming Lin **Fangjun Li**

School of Computer Science, University of Manchester

Project Page: `https://fmlinks.github.io/library/generative-ai/`

ABSTRACT

Generative artificial intelligence (generative AI) in computer vision and medical image analysis encompasses different tasks, model families, and validation requirements. This survey connects machine learning foundations, visual generation, and medical imaging through eight composable dimensions and 27 subtasks. We explain image synthesis, image-to-image translation, image editing, restoration and inpainting, inverse problems and image reconstruction, and video generation. The taxonomy relates conditional generation, text-to-image generation and controllability to few-shot, zero-shot and data-efficient learning, multitask and multimodal generation, joint outputs, and image, texture and shape representations. It also distinguishes 4D generation, frame interpolation and action-conditioned world models, and compares generative adversarial networks (GANs), diffusion models and flow matching. Fidelity, efficiency and measurement consistency provide task-specific evaluation criteria. A fixed-query analysis of 25,056 arXiv title-abstract records from 2018–2026 year-to-date characterizes vocabulary trends and co-occurrence, complemented by a purposive synthesis of 60 source families and 27 timelines covering 383 distinct publications. The analysis reveals the historical shift from GAN to diffusion vocabulary and growing action-related language; these are literature signals, not method-adoption rates or clinical evidence. Medical use cases and validation needs include synthetic CT, MRI reconstruction, virtual staining, anatomical shape and motion modeling, and synthetic cohorts for in silico trials. The synthesis emphasizes context of use, fair comparison, independent references and explicit evidence boundaries, providing general considerations for interpreting published results. This framework supports generative-model selection, evaluation design, and evidence interpretation beyond visual realism.

***K*eywords** Generative artificial intelligence · Computer vision · Medical image analysis · Conditional generation · Diffusion models · Flow matching · Image reconstruction · In silico trials

1 Scope and Methods

The object of this survey is visual generation, not every use of generative AI. We connect machine-learning formulations to CV tasks and medical-image applications, including synthesis, translation, editing, restoration, measurement reconstruction and temporal generation. Text appears as a condition or paired output; general language, audio and code generation are outside the central scope. Existing vision and medical surveys and the structural reference offer complementary coverage [1–3]. Our purpose is to explain what each task means and what evidence would make a next study useful, rather than to catalogue model names or forecast acceptance at a venue.

1.1 Discovery, Quantitative Frame, and Evidence Selection

Three collections must not be conflated. The multi-source discovery archive contains 128,499 source records, 124,215 conservatively deduplicated families and 58,516 automatic candidates; these are not manually included papers. The quantitative analysis instead uses one frozen visual-task arXiv query: 18,037 unique IDs from 2018–2025 in 35 hash-checked pages and 7,019 from 2026 in eight further pages, totaling $N = 25,056$. The snapshot finished on 21 October 2026 at 15:00 UTC; returned 2026 publication dates end on 20 October. Category and phrase restrictions,

indexing lag and arXiv participation limit coverage. Separately, 60 purposively selected source families support the qualitative evidence pass, spanning ML foundations, CV methods, medical neighbors, clinical studies and guidance. None of these collections represents complete journal or conference coverage.

Inspection depth remains explicit. Selected full-text methods, results or limitations passages were checked for 54 of the 60 families. Two reviews received scope inspection, the user-provided reference structural inspection, one clinical MRI source indexed primary-section inspection, one FDA source official-scope inspection, and one staining neighbor abstract-only inspection. This is not cover-to-cover reading of every source. Targeted searches combined task synonyms with validation, cohort, motion, clinical and control terms, followed by direct-neighbor and contrary-evidence checks. Versions and retrieval hashes remain separate even when arXiv/journal aliases are merged. Additional correspondence examples have their own source/depth log; the extraction has one Codex-assisted reviewer, not independent clinical adjudication.

1.2 Coding, Trajectories, and Sensitivity

The relevance audit argues against a filtered-corpus field share. The historical 102-rule system retains 12,237 and rejects 5,800 of 18,037 records. With seed 20261021, 24 titles and complete abstracts were inspected in each early/late period by retained/rejected stratum. Relevant counts were 19/24 and 20/24 among retained records, and 6/24 and 12/24 among rejected records, with one uncertain early rejection. These conditional counts reveal both contamination and missed work; they are not a gold-standard accuracy estimate. Sensitivity compares the full query frame, rule-retained records, title-only coding and exclusion of review titles for complete years 2018–2025. No separate 2026 relevance audit or per-label precision/recall validation is claimed.

Figure trajectories measure weighted vocabulary, while headings count matches. Let b_{ij} denote a binary title–abstract match for concept j. Freeze historical weights as

$$d_j = 1 + \ln \frac{18,037 + 1}{\mathrm{df}_{j,2018:2025} + 1}, \quad z_i = \frac{b_i d}{\|b_i d\|_2}, \quad I_{jy} = \frac{1}{N_y} \sum_{i:\,\mathrm{year}(i)=y} z_{ij}, \tag{1}$$

with $z_i = 0$ for unmatched records. All 102 concepts participate in normalization, not only the 27 displayed ones. Applying the same IDF to 2026 preserves historical weights. Annual denominators include all queried records, not just matches. Dashed terminal segments and open markers indicate 2026 YTD, without annualization. Co-mention analysis in Section 4 reports its own conditional denominators; it is not method adoption, causal dependence or a matched-calendar growth estimate.

1.3 Representative Histories and Reading Order

The histories establish landmarks, not a global citation Top 20. Each of the 27 subitems contains 20 different representative papers chosen for methodological relevance, capability coverage and recent developments: 540 memberships, 383 distinct publications. These are metadata/title–abstract checks, not 383 additional full-text reviews. Dates use verified arXiv v1 submissions where available, with exceptions logged individually; an earlier project or OpenReview release is not ruled out. Timelines cover 2018–2026*, through 22 October.

The presentation proceeds from the complete eight-coordinate overview to all 27 explanations and histories in coordinate order. Only then do we synthesize cross-dimensional evidence, examine selected medical needs and discuss general evidence considerations. Correspondence diagrams describe shared formulations and illustrative applications, not a measured ML-to-clinic transfer pipeline. Their Need/Test statements specify evaluation requirements, not achievements attributed to every example.

2 Overview: Eight Coordinates and 27 Subitems

Read a model name as a combination of coordinates. Figure 1 aligns eight first-level groups with 27 numbered subitems. For example, few-shot text-conditioned texture synthesis combines a data regime, condition, object and operation; these are not four competing tasks. The atlas retains one pooled trajectory per panel and introduces the vocabulary expanded in the following section.

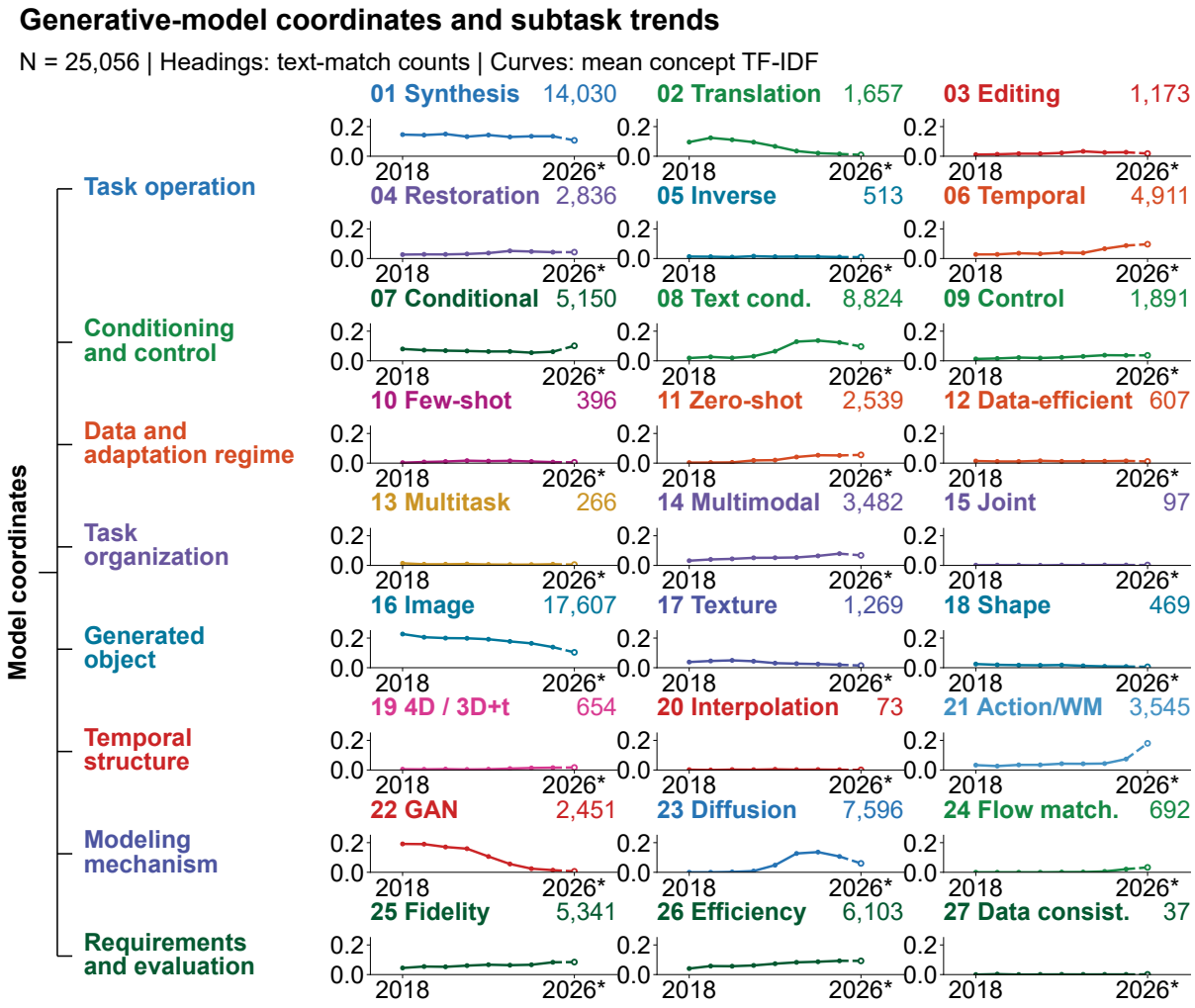


Figure 1: **Generative-model coordinates and subtask trajectories.** Eight groups, 27 selected subitems. Counts are overlapping title–abstract matches in $N = 25,056$ records, including 7,019 from 2026 YTD. Curves use Equation 1 and common 0–0.25 limits. Asterisks, open endpoints and dashed segments mark incomplete 2026, retrieved 21 October. WM denotes world model.

The first-level groups ask different questions. Operation specifies what changes; conditioning specifies information supplied; data regime specifies the target-data budget; organization specifies relationships between tasks or outputs; object specifies what is generated; temporal structure specifies dependence across time or action; mechanism specifies how a distribution is learned or sampled; requirements specify what must be preserved and tested. The 27 leaves are selected vocabulary anchors, not an exhaustive partition. A paper can occupy several leaves and most useful experiments cross several coordinates. Table 1 makes the interpretation contract explicit without repeating the atlas counts.

The coordinate contract identifies which assumption changes when two methods are compared. Its final column names common category errors that otherwise make apparent progress difficult to interpret.

Table 1: **Eight-coordinate reading contract.** Scope: the survey's complete taxonomy; groups are composable, not exclusive.

Coordinate	Question to answer	Do not infer
Task operation	What is created, changed or recovered?	Every plausible sample recovers an observed case.
Conditioning and control	Which input is supplied, and what can be steered?	Conditioning guarantees selective control.
Data and adaptation regime	Which real examples and pretrained resources are allowed?	Few target samples mean little total training.
Task organization	What shares learning or forms a coupled sample?	Multimodal, multitask and joint are synonyms.
Generated object	Intensity, appearance or geometry?	A 3D intensity volume is a surface model.
Temporal structure	What depends on time, history or action?	Smoothness establishes valid dynamics.
Modeling mechanism	Which objective, probability path and sampler?	Backbone choice defines the task.
Requirements and evaluation	Which quantity, budget and failure must be checked?	An attractive mean score certifies every use.

Distributional and measurement conditions imply different obligations. Unconditional sampling models $p_\theta(x)$ and semantic conditioning models $p_\theta(x \mid c)$. A particular measurement instead constrains a particular object:

$$y = A(x) + \epsilon, \qquad p(x \mid y, A) \propto p(y \mid x, A)p(x). \tag{2}$$

A prompt admits many suitable images; patient data refer to one underlying case. The operator, noise and prior all influence a reconstruction. A deterministic encoder–decoder need not define a sampling distribution, while a variational latent model does so explicitly [4]. Restricting solutions to a generator's range can also introduce representation error [5]. Probability notation describes the intended target; it does not certify a practical sampler.

A coordinate must not stand in for its missing neighbors. Few-shot target adaptation may depend on abundant source pretraining [6]. Texture may mean surface appearance on supplied geometry, tissue appearance, or a quantitative imaging signature. A Transformer is a backbone, not by itself a distributional objective; a dynamic output is not necessarily an action model. These distinctions prevent a change of representation or terminology from being mistaken for a new task. The next section follows items 01–27 exactly, with a correspondence map before each group's individual histories.

3 Eight Dimensions in Detail

Each group begins with a shared ML–CV–medical map, followed by its numbered subitems. The maps supply examples and invariants; the histories supply dated landmarks. All 27 items are introduced here before any cross-dimensional synthesis or general evidence discussion.

3.1 Task operation

An operation says what is created, changed or recovered. The six rows separate new-instance sampling from identity-preserving translation, local edits, missing-content recovery, measurement inversion and sequence generation. Their conditions and acceptable errors differ even when one model family supports several operations.

Figure 2 gives the examples for this coordinate of Figure 1.

01 Task operation

Figure 1 coordinate 1 | Subitems 01, 02, 03, 04, 05, 06

Machine learning	Computer vision	Medical image analysis
01 Synthesis	**Image / texture / shape**	**MAISI synthetic CT**
Noise + condition -> new sample Learn a sample distribution Keep requested properties Test coverage and diversity	Noise / prompt -> visual sample LDM; Text2Tex Keep requested attributes Test fidelity and coverage	Masks / metadata -> CT volume Vary appearance, retain anatomy Need independent and privacy QA Test held-out real-task utility
02 Translation	**Image-to-image translation**	**Synthetic CT / virtual stain**
Source domain -> target domain Learn a conditional mapping Keep relevant subject identity Pairing changes assumptions	Source -> target appearance pix2pix; CycleGAN Keep scene / object structure Test subject-wise preservation	MRI -> CT; tissue -> stain MR-only RT; virtual histology Need subject-specific fidelity Test dose / pathology and QA
03 Editing	**Instruction editing**	**Anatomic digital siblings**
Image + edit -> modified image Change a specified property Keep non-target content Not a treatment counterfactual	Image + instruction -> edit InstructPix2Pix Keep identity outside the edit Test target / off-target change	Local edit -> related anatomy Diffusion-based local editing Need controlled variation Test local truth and invariants
04 Restoration / fill	**Denoise / enhance / inpaint**	**Microscopy / region filling**
Degraded input -> restored output Recover or fill missing content Keep observed information Separate truth from completion	Partial image -> restored image Palette; DDRM Keep supported detail Test false structures	Degraded to enhanced content Microscopy; MAISI tumor fill Separate recovery / simulation Test hallucination + task error
05 Measurement inverse	**Inverse reconstruction**	**MRI / sparse-view CT**
Data + operator -> reconstruction Infer a constrained posterior Unobserved parts stay uncertain Noise assumptions matter	Known operator -> image draws Diffusion posterior sampling Keep likelihood-supported detail Test operator shift / calibration	k-space / projections -> image Score-based medical inversion Need patient / lesion fidelity Test data shifts / decision error
06 Temporal generation	**Video / 4D / world models**	**Cardiac cycles / ultrasound**
Past / action -> future states Model dependent temporal states Keep meaningful dynamics Actions are optional conditions	History / control -> future view Video diffusion; Genie; 4D-fy Keep temporal / action relations Test unseen actions	Phase / pose -> anatomy / view CHeart; Mesh Flow; SonoWorld Need local motion / action truth Test against static / replay

Figure 2: **Task operation: ML–CV–medical correspondences.** Figure 1, items 01–06. Examples and evaluation requirements; see Section 1. Sources: [7–26].

3.1.1 01 Synthesis

Synthesis samples an instance rather than recovers a specific observed person. A text-to-image model creates a scene compatible with a description; a medical volume generator creates a possible anatomy under supplied conditions. The input can be noise alone or a combination of noise and metadata, while the output may be pixels, voxels, geometry or texture. Latent diffusion is a representative visual implementation [7]. In medicine, MAISI conditions volumetric generation on body region and spacing and provides mask-guided or tumor-filling paths [9]. Texture is another object coordinate: Text2Tex generates surface appearance on an already supplied mesh, so it does not demonstrate joint geometry recovery [8]. The elementary question is which attributes the conditions control and which variations remain free, not simply whether the rendered result looks realistic.

Synthesis is useful when its variation supports a real downstream need. Training augmentation requires improvement on held-out real cases, not performance on generated labels alone; cohort construction requires representative coverage, not sample count alone. MAISI includes external autoPET image-distribution evaluation, but that is not an independent diagnostic endpoint or privacy audit [9]. MAISI-v2 combines rectified flow, region-specific losses and changed data/training conditions, so differences cannot all be attributed to one objective [27]. Conditional anatomical modeling also has existing implementations [28, 29]. A responsible comparison measures conditional accuracy, evaluation coverage, memorization risk and real-task usefulness at matched resources. If resampling or a statistical shape model already supplies the required variation, a larger learned generator must justify its extra assumptions and costs.

Figure 3 places 20 representative synthesis publications in the common 2018–2026* window; the arrows resolve to the bibliography [7, 30–48]. The common selection and date conventions are defined in Section 1.

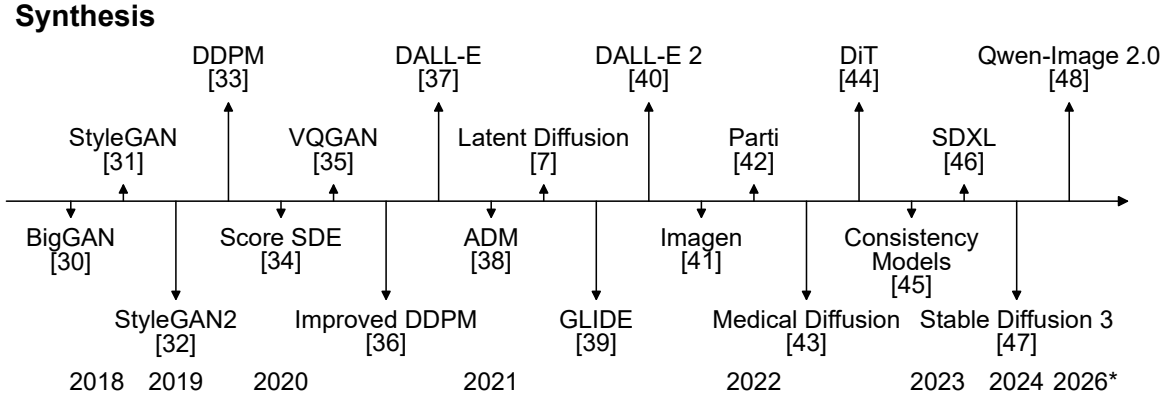


Figure 3: **01 Synthesis: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.1.2 02 Translation

Translation changes the domain while trying to preserve the relevant object. Turning a label map into a photograph changes appearance; turning MRI into synthetic CT changes an imaging representation; virtual staining maps an unstained measurement to stain-like contrast. Paired translation learns from aligned examples, whereas unpaired translation learns from marginal domain samples with additional constraints [10, 11]. The distinction is consequential: cycle consistency can support an invertible mapping without uniquely identifying the medically correct subject-wise mapping. Virtual staining demonstrates a genuine medical translation task, with registration and tissue preparation affecting the reference pairs [13]. The output should therefore be described as a conditional estimate under a specific acquisition and training protocol, not a new measurement of every biological quantity that the target modality might reveal.

The preservation contract determines the evaluation. For synthetic CT, density and geometric errors matter through dose calculation and positioning, while for staining they matter through structures and diagnostic observations. AortaDiff already couples contrast translation with segmentation and clinically interpretable measurements, so "task-aware joint translation" is not an untouched idea [49]. Its preprint evaluation also has ROI, alignment and derived-reference limitations that should remain visible. The SynthRAD2025 report separates image, geometric and dose criteria and demonstrates why rankings depend on the endpoint [50]. Unpaired realism or average similarity is insufficient when a clinically relevant structure changes treatment planning. Translation should be compared against the current clinical workflow, including non-generative alternatives, and failures should trigger an explicit fallback rather than be hidden in a favorable mean score.

Figure 4 places 20 representative translation publications in the common 2018–2026* window; the arrows resolve to the bibliography [51–70]. The common selection and date conventions are defined in Section 1.

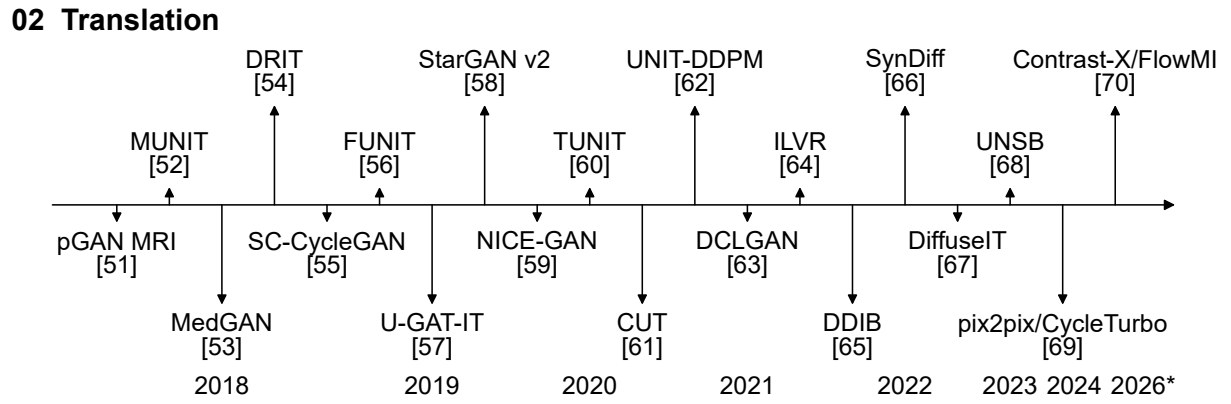


Figure 4: **02 Translation: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.1.3 03 Editing

Editing asks for a controlled difference from an existing instance. An instruction such as changing an object's color is not unconditional sampling; the original image supplies identity and non-target attributes. InstructPix2Pix learns from generated instruction-edit triplets and reports difficulties with unwanted changes, isolation and viewpoints [14]. In medicine, digital siblings already create local anatomic variations and assess morphology and topology [15]. Here, the useful output is often a family of controlled alternatives for sensitivity analysis rather than a clinically asserted future patient. One must specify the editable region, required invariants, permitted topology changes and the meanings of edit magnitude. A visually small change can still be large for a downstream measurement, especially near a lesion, valve or vessel boundary.

A counterfactual image is not automatically a causal treatment counterfactual. An edited anatomy can support a computational experiment if the simulator and intervention are explicitly defined, but visual plausibility alone does not identify how a patient would respond to treatment. Distributional editing also cannot establish that all non-target physiology is preserved. Appropriate tests compare independent measurements before and after editing, stress boundary cases, and quantify collateral changes outside the requested region. Classical parameterized deformation or resampling can be a strong comparator when the desired variation is simple. The contribution of a learned editor should be a newly supported, validated class of variation or a better error–cost trade-off, not the relabeling of existing anatomy manipulation. This distinction keeps controlled sensitivity studies useful without overstating biological causality.

Figure 5 places 20 representative editing publications in the common 2018–2026* window; the arrows resolve to the bibliography [14, 48, 71–88]. The common selection and date conventions are defined in Section 1.

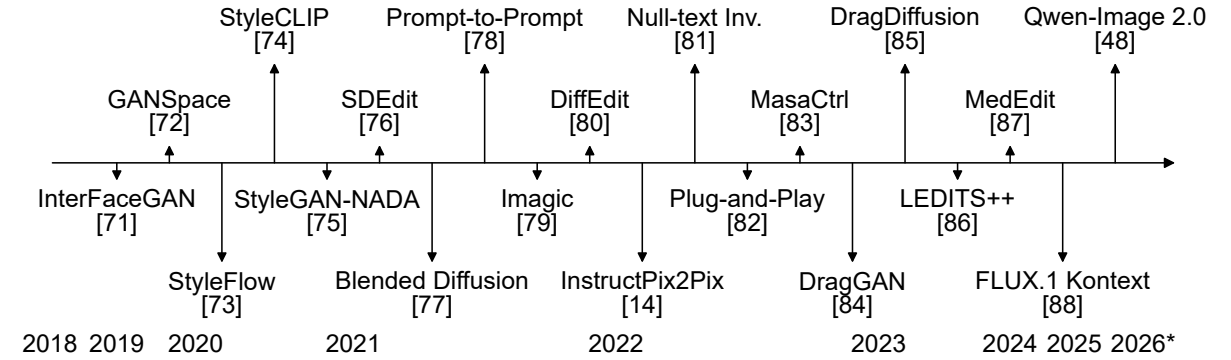


Figure 5: **03 Editing: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.1.4 04 Restoration / fill

Restoration retains an observed scene while compensating for degradation. Denoising, deblurring, colorization and filling missing regions differ in how much of the answer is identifiable. Palette uses a shared image-conditioned diffusion formulation across several such tasks [16]; multiple conditions do not alone establish multitask learning without shared task training and evaluation. DDRM reuses an image prior across linear inverse problems through operator structure [17]. Medical examples include microscopy restoration and MAISI tumor filling [9, 18]. A filled region may be a plausible simulation object rather than evidence of the true hidden anatomy. The task description must say whether the aim is retrospective recovery, prospective diagnosis, synthetic augmentation or experimental perturbation, because identical-looking outputs can have very different acceptable failure modes.

The unobserved region is where a strong prior can become a liability. A small residual on visible pixels says little about missing structures; several plausible samples do not automatically quantify calibrated uncertainty. Existing work already separates prior-induced hallucinations from ordinary reconstruction error and develops distribution-referenced hallucination metrics [18, 89]. The latter depends on a defined reference and approximation assumptions, not a universal truth oracle. Evaluation should therefore distinguish measurement-supported detail, ambiguous completion and wrong but convincing structures. For a clinical use, the question is whether the ambiguity changes a diagnosis or quantitative endpoint and whether the system can decline to fill it. Deterministic restoration remains an essential comparator: this operation-based taxonomy includes such comparators without claiming that every denoiser is a stochastic generative model.

Figure 6 places 20 representative restoration / fill publications in the common 2018–2026* window; the arrows resolve to the bibliography [16, 90–108]. The common selection and date conventions are defined in Section 1.

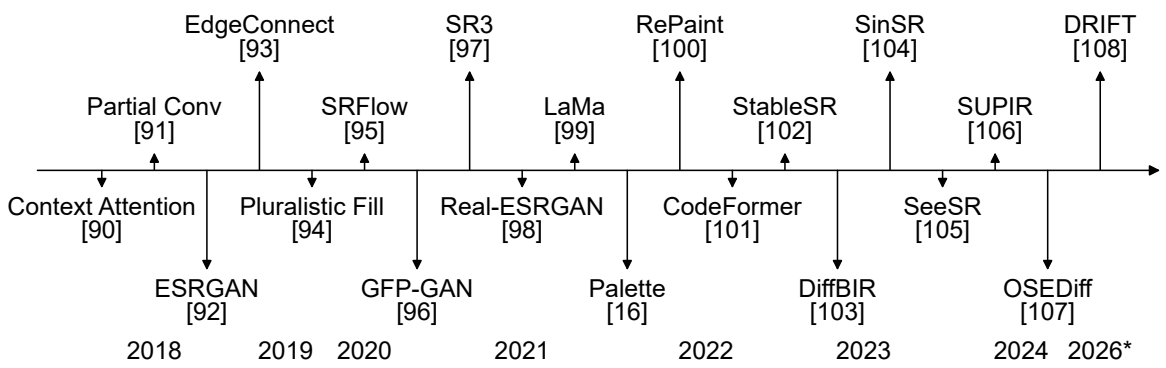


Figure 6: **04 Restoration / fill: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.1.5 05 Measurement inverse

Reconstruction uses a specified forward model and a particular measurement. CT projection data and MRI k-space constrain a latent object through Equation 2; they are not merely prompts for a similar-looking image. Generative compressed sensing makes representation error explicit, while DPS approximates a noisy likelihood using denoised estimates [5, 19]. Medical score-based reconstruction demonstrates reuse of an unconditional prior under different sampling operators [20]. In the inspected study, CT projections and single-coil MRI measurements are simulated from image data. Those experiments probe operator variation, but they do not constitute prospective scanner or diagnostic validation. A precise report separates real source images, simulated measurements, real acquisition tests, external patients and clinical readers rather than collapsing all of them into "real medical data."

Faithfulness requires more than exact data fitting. Noise and model mismatch can make strict fitting undesirable, while a small data residual cannot certify components poorly constrained by the operator [89]. Generator-range bias can persist even with more observations [5], and posterior-sample diversity can mix model, numerical and observation effects. A stronger experiment fixes the raw-data split, compares a competitive non-generative method and a generative baseline, and measures a prespecified lesion, quantitative or reader endpoint at a realistic compute budget. Active acquisition is already explored with diffusion posterior samples [109]; novelty must lie beyond merely choosing the next view. Uncertainty becomes practically useful only when it supports a calibrated decision, rejection or additional measurement with a favorable total burden, including failed cases and reacquisition time.

Figure 7 places 20 representative measurement inverse publications in the common 2018–2026* window; the arrows resolve to the bibliography [17, 19, 20, 110–126]. The common selection and date conventions are defined in Section 1.

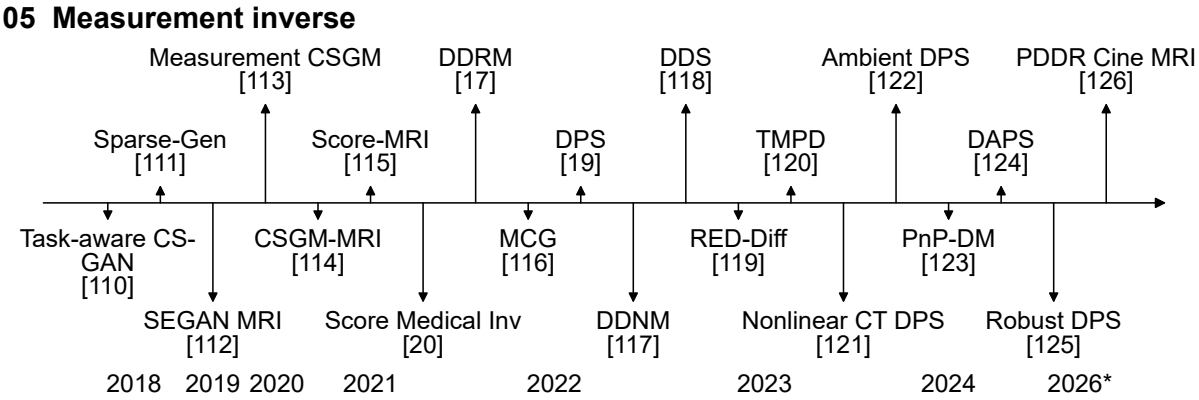


Figure 7: **05 Measurement inverse: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.1.6 06 Temporal generation

Time introduces several distinct problems. Unconditional video creates a sequence; interpolation fills missing times; prediction conditions on a past; action-conditioned generation also receives an intervention on the acquisition or environment. Video diffusion and Genie exemplify joint visual sequences and interactive dynamics [21, 22]; 4D-fy adds a dynamic 3D representation with complementary image and video supervision [23]. Medical cases include CHeart's within-cycle anatomy, Cardiac Mesh Flow's dynamic meshes and ultrasound views under probe movement [24–26]. A cardiac cycle is not longitudinal disease progression, and an autoregressive "causal" architecture is not a treatment-effect model. Similarly, a smooth rendered mesh is not automatically a measured displacement field. Each temporal use must specify its conditioning history, horizon, action semantics and independent reference.

Temporal validity needs negative controls that defeat superficial metrics. Cardiac Mesh Flow's ablation shows that static hearts can attain strong periodic consistency, making endpoint agreement an insufficient motion validator [25]. CHeart's dense temporal anatomy relies on processed segmentation labels, not independent local motion truth [24]. SonoWorld already grounds ultrasound generation in metric poses, while EchoWorld studies predictive representations for probe guidance [26, 127]; neither inspected evaluation establishes an independent real-robot clinical policy-transfer trial. Contact force, deformation, physiology and scanner settings can all break a pose-only observation model. Tests should include static outputs, condition shuffling, non-replayed actions and local motion or control endpoints. The important advance is not longer or smoother video, but a demonstrably correct relation between time, action and the downstream quantity.

Figure 8 places 20 representative temporal generation publications in the common 2018–2026* window; the arrows resolve to the bibliography [21, 128–146]. The common selection and date conventions are defined in Section 1.

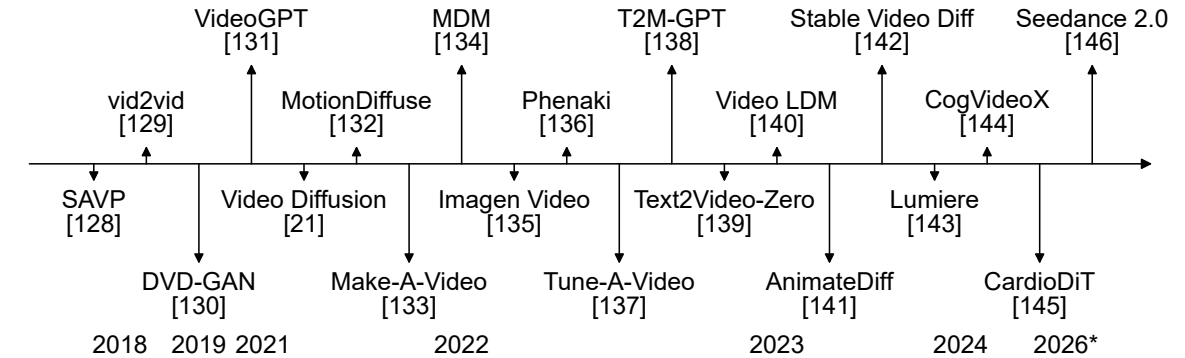


Figure 8: **06 Temporal generation: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.2 Conditioning and control

Conditions specify supplied information; control asks whether a requested property changes reliably. Text is one condition, not a synonym for conditioning. A model can use masks or words while still failing to preserve unrequested content. These distinctions explain the three rows below.

Figure 9 gives the examples for this coordinate of Figure 1.

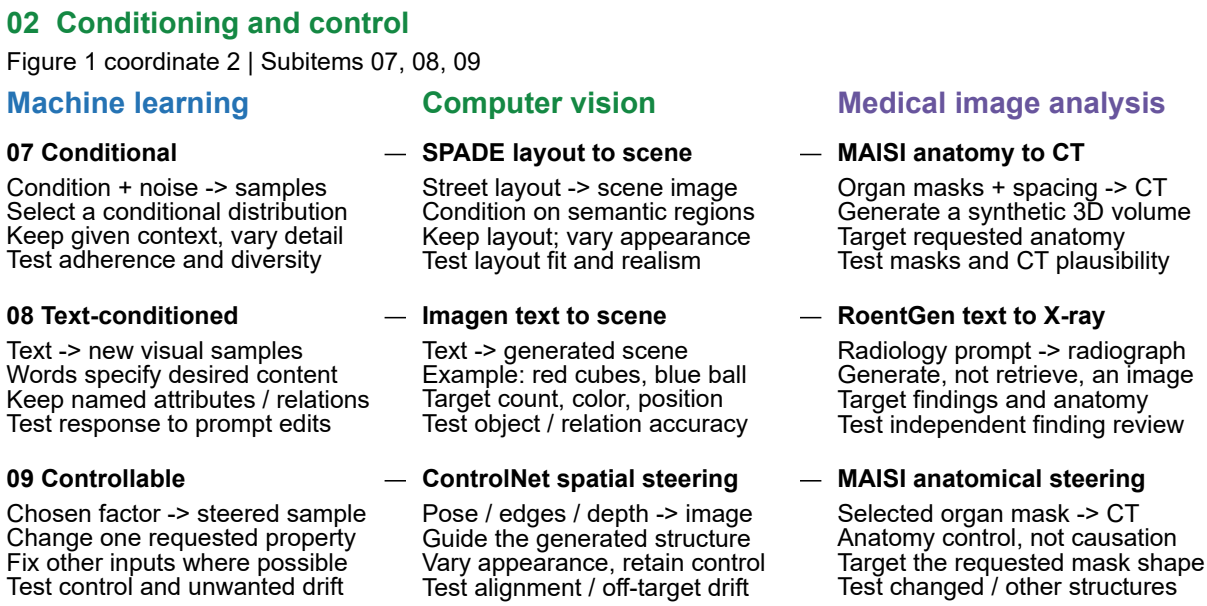


Figure 9: **Conditioning and control: ML–CV–medical correspondences.** Figure 1, items 07–09. Examples and evaluation requirements; see Section 1. Sources: [9, 41, 147–149].

3.2.1 07 Conditional

Conditional. Conditional generation asks how outputs change when information is supplied, not how much training data are available. In ML, a condition selects a distribution: the same class or context can still admit many valid samples. SPADE transfers this idea to CV by synthesizing scenes from semantic layouts. MAISI provides a medical counterpart, generating synthetic three-dimensional CT with organ segmentations as conditions and configurable voxel spacing. The desired invariant is agreement with the supplied anatomy; unsupplied appearance remains variable. Text conditioning is one special case, while controllability additionally asks whether specific changes can be steered reliably. Suggested checks are condition agreement, diversity and anatomical plausibility, not assumed guarantees. A conditioning input does not establish patient identity, pathology truth, privacy or clinical benefit. The timeline and examples draw on the following sources [7, 9, 16, 30, 37, 39, 41, 52, 97, 147–157].

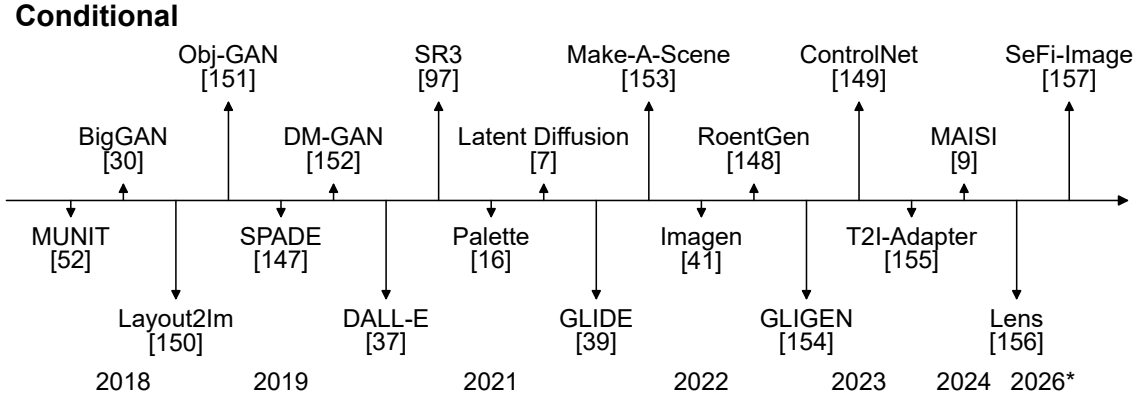


Figure 10: **07 Conditional: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.2.2 08 Text-conditioned

Text-conditioned. Text-conditioned generation uses language as the condition; it is therefore compatible with other conditioning modalities and data regimes. In ML, words specify requested attributes and relations, but the generator must translate them into visual content. Imagen provides a CV anchor for text-to-image diffusion, making object counts, colors and relations useful teaching probes rather than guaranteed successes. RoentGen adapts a pretrained latent diffusion model using chest radiographs and reports, then generates radiographs from medical-language prompts. Suggested checks distinguish prompt adherence from image realism and independently assess depicted findings and anatomy. A requested diagnosis is neither evidence that the output depicts it correctly nor truth about a patient. These abstract-supported examples establish generation capabilities, not complete pathology control, patient preservation or clinical proof. The timeline and examples draw on the following sources [7, 37, 39–42, 46, 47, 148, 151–153, 156–163].

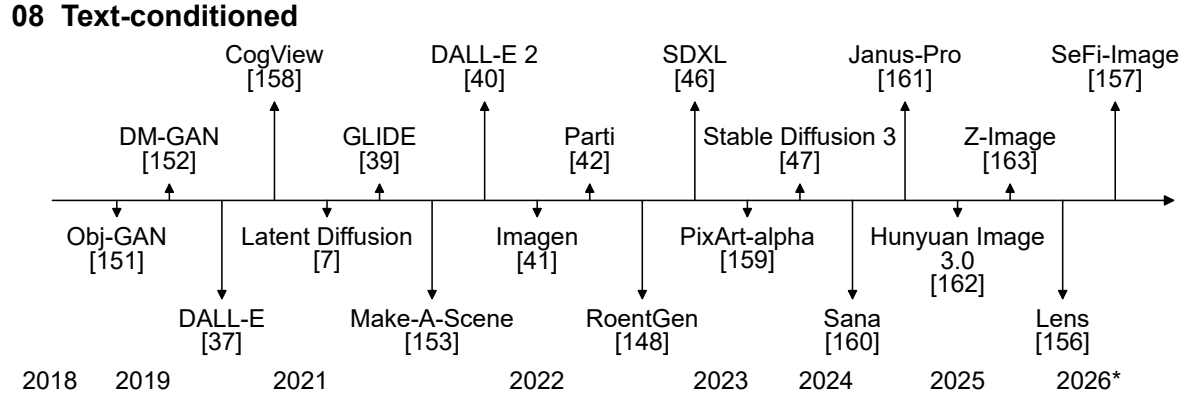


Figure 11: **08 Text-conditioned: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.2.3 09 Controllable

Controllable. Controllability is a steering objective, whereas conditioning describes information supplied to a generator. A useful ML probe varies one requested factor while fixing other inputs and, when meaningful, the random seed; outputs must still be checked for coupled changes. ControlNet adds trained spatial controllers to a pretrained text-to-image diffusion model using pose, edge or depth conditions. MAISI documents organ-mask conditioning for CT synthesis, providing a narrower anatomical-control example rather than a disease-intervention model. Suggested checks quantify response to the changed mask and unintended changes elsewhere. Neither a frozen base generator nor an anatomical control implies training-free control learning, causal identification, preserved patient identity or clinical validity. These coordinates can overlap with text conditioning and with either low-data adaptation or extensive training. The timeline and examples draw on the following sources [9, 14, 71–74, 78, 83, 147, 149, 153–155, 164–170].

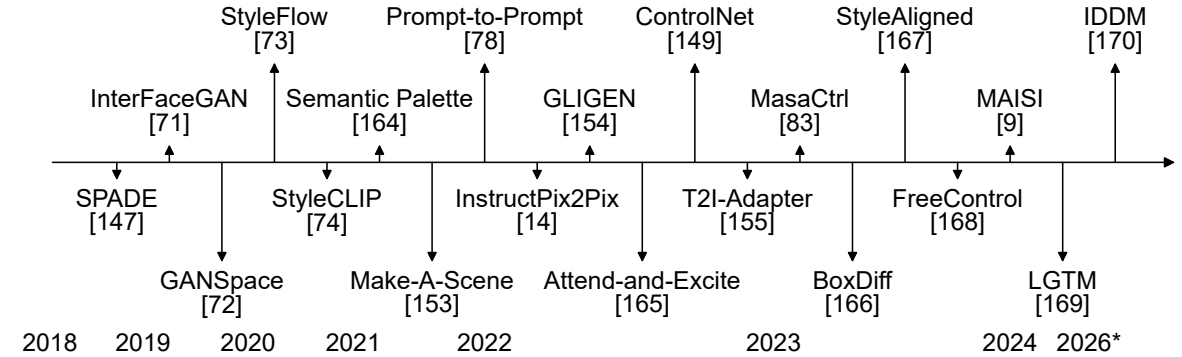


Figure 12: **09 Controllable: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.3 Data and adaptation regime

The budget must identify target examples, source pretraining and inference-time fitting separately. Few-shot limits a target support set; zero-shot or training-free reuse removes a specified fitting stage; data efficiency compares performance against a real-data budget. None alone establishes low total computation.

Figure 13 gives the examples for this coordinate of Figure 1.

03 Data and adaptation regime

Figure 1 coordinate 3 | Subitems 10, 11, 12

Machine learning	Computer vision	Medical image analysis
10 Few-shot	**DreamBooth personalization**	**EndoFSA endoscopy**
Few examples -> adapted model Reuse a learned source prior Keep target traits, vary samples Test novelty and memorization	Few photos -> new subject views Fine-tune a pretrained generator Target identity, not copied views Test identity / novelty / prompts	Few lesion images -> adaptation Reuse normal-image pretraining Generate abnormal endoscopy Test lesions and memorization
11 Zero-shot / training-free	**RePaint without retraining**	**Score-MRI sampling reuse**
New condition -> prior reuse No target generator refitting Pretraining can still be large Specify what receives no training	Image + mask -> filled image Reuse an unconditional DDPM Modify sampling, not the network Test visible pixels / filled region	Sparse MRI data -> image Reuse a magnitude-MRI prior No sampling-pattern refitting Test data fit and missed lesions
12 Data-efficient	**ADA limited-data GAN**	**EndoFSA target-data reuse**
Limited real data -> generator Reduce the real-example budget Keep variation, avoid overfitting Test quality versus data count	Limited images -> trained GAN Adapt critic-side augmentation Compare at equal data budgets Test diversity and sample quality	Scarce lesions -> adapted model Reuse a normal-data prior Keep pretraining budget explicit Test real-patient utility / budget

Figure 13: **Data and adaptation regime: ML–CV–medical correspondences.** Figure 1, items 10–12. Examples and evaluation requirements; see Section 1. Sources: [100, 115, 171–173].

3.3.1 10 Few-shot

Few-shot. Few-shot generation limits examples of a specified target, such as a subject or domain, rather than all data ever used. In ML, adaptation transfers a learned prior to that small support set; other methods can condition on support without weight updates. DreamBooth illustrates CV personalization by fine-tuning a pretrained text-to-image model on a few subject photographs. EndoFSA documents a medical case: a generator pretrained on abundant normal capsule-endoscopy images is adapted to scarce abnormal domains through restricted parameter updates. These are generated domain examples, not reconstructions of a particular patient's lesion. Suggested checks include unseen-patient lesion review, diversity and memorization auditing. Exact medical shot counts and independent clinical performance require further evidence; few trainable parameters alone would not establish few-shot learning. The timeline and examples draw on the following sources [6, 170–172, 174–189].

10 Few-shot

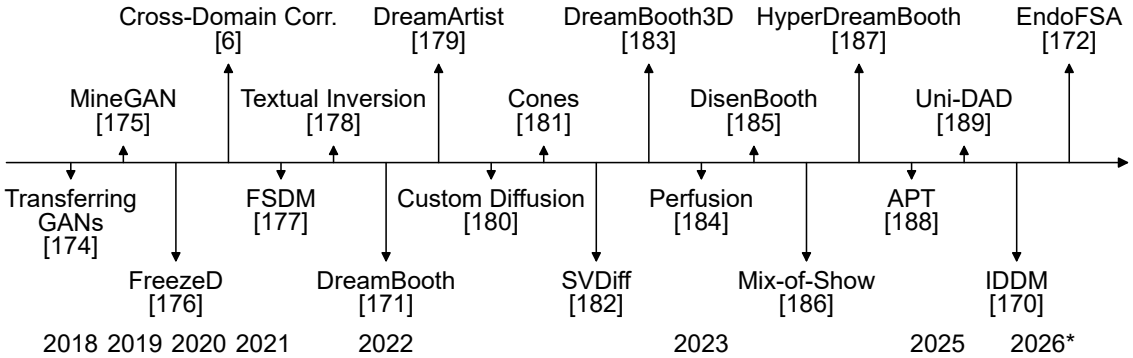


Figure 14: **10 Few-shot: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.3.2 11 Zero-shot / training-free

Zero-shot / training-free. This coordinate concerns reuse without fitting generator weights to the target condition or acquisition operator. It differs from semantic zero-shot recognition of unseen categories. No target labels also does not mean no pretraining or no target-modality images. In ML, an existing prior is combined with a new condition during inference. RePaint turns a trained unconditional diffusion model into an inpainter by changing reverse-sampling iterations rather than retraining its network. In medicine, Score-MRI combines a magnitude-image-trained score prior with data-consistency steps across sampling patterns. Its training-free aspect is acquisition-specific reuse, not learning MRI anatomy from nothing. Suggested tests assess measurement residuals and independently checked anatomy. Inference projections or latent optimization still incur computation; agreement with measured data cannot verify unmeasured lesions. The timeline and examples draw on the following sources [17, 19, 64, 76, 78, 80, 82, 83, 100, 115, 117, 139, 165–169, 190–192].

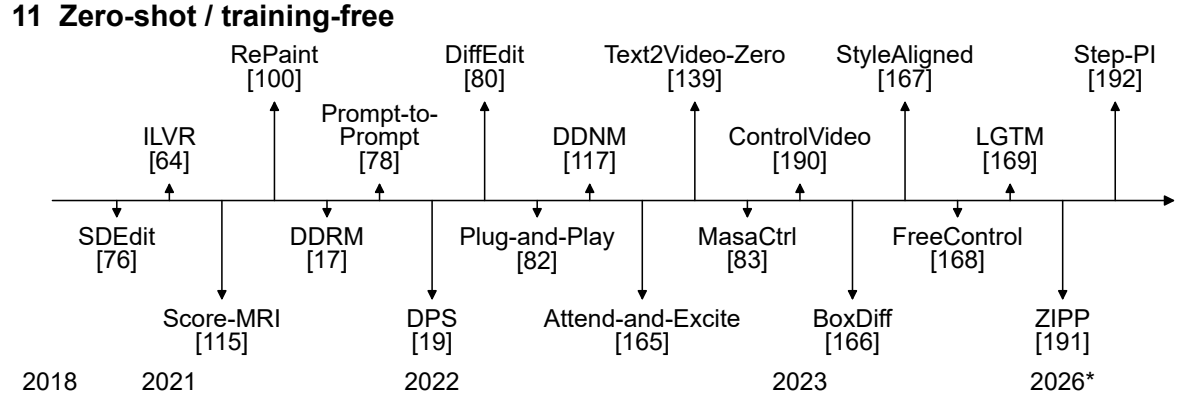


Figure 15: **11 Zero-shot / training-free: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.3.3 12 Data-efficient

Data-efficient. Data efficiency asks whether useful generation can be learned with fewer real examples or less target supervision at a specified quality level. In ML, regularization or transfer can reduce overfitting, but the data budget must be stated. ADA provides a CV anchor through adaptive discriminator augmentation for limited-data GAN training, including training from scratch and transfer. EndoFSA illustrates medical target-data reuse: normal-image pretraining supports adaptation to scarce abnormal endoscopy images. This is a documented low-data design, not a verified cross-method efficiency ranking. Suggested tests plot quality, diversity and independent downstream performance against real patient counts with fixed splits and comparable pretraining access. Count abundant pretraining separately; faster sampling, smaller adapters and more synthetic images do not themselves demonstrate lower real-data requirements. The timeline and examples draw on the following sources [6, 171–178, 180, 193–202].

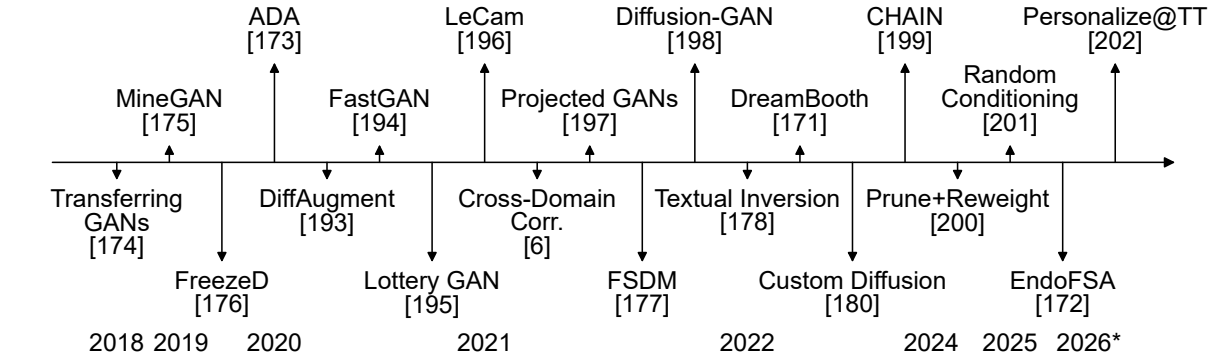


Figure 16: **12 Data-efficient: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.4 Task organization

Organization describes what shares learning and what belongs to the same sample. Multitask concerns objectives, multimodal concerns representations, and joint generation concerns dependent outputs. The same system can have all three properties; architecture or channel count alone establishes none of them.

Figure 17 gives the examples for this coordinate of Figure 1.

04 Task organization

Figure 1 coordinate 4 | Subitems 13, 14, 15

Machine learning	Computer vision	Medical image analysis
13 Multitask	— **Palette generalist model**	— **AortaDiff shared objectives**
Several tasks -> shared model Share learning across objectives Keep separate task evaluations Test each task versus specialists	Task + input -> restored image Colorize, inpaint, uncrop, restore Share conditional diffusion Test each task, not their average	Plain CT -> synthetic CT / masks Synthesis + segmentation tasks Target shared anatomical identity Test image / Dice / size error
14 Multimodal	— **UniDiffuser image and text**	— **UniX reports and X-rays**
Different modalities -> modeling Link language and visual content Conditioning is not joint sampling Test cross-modal agreement	Text -> image; image -> text Also supports paired sampling Specify the sampling mode Test alignment in each mode	Report -> X-ray; X-ray -> report Generate across distinct formats Target findings, not patient ID Test omissions / contradictions
15 Joint generation	— **JointNet RGB and depth**	— **MedSegFactory image / mask**
Noise -> corresponding outputs Learn how outputs depend Describe one shared sample Test pairs and individual outputs	Text + noise -> RGB-depth pair Couple two diffusion branches Target matching scene geometry Test edges and pair consistency	Text + noise -> image-mask pair Cross-condition two streams Target matching anatomy / labels Test boundaries / real-task utility

Figure 17: **Task organization: ML–CV–medical correspondences.** Figure 1, items 13–15. Examples and evaluation requirements; see Section 1. Sources: [16, 49, 203–206].

3.4.1 13 Multitask

Multitask. Multitask describes how objectives share learning, not how many channels enter a network. A generic ML example shares parameters across conditional generation objectives while retaining task-specific evaluation under shared or task-specific losses. Palette provides a CV instance: one generalist diffusion model handles colorization, inpainting, uncropping, and JPEG restoration. In medicine, AortaDiff combines synthetic contrast-enhanced CT generation from noncontrast CT with lumen and thrombus segmentation. It therefore also overlaps joint generation; coordinate values are not exclusive categories. Both image quality and segmentation or measurement error require separate checks against specialist baselines. The intended invariant is anatomical identity across tasks, not identical pixel values. Shared architecture alone does not establish a single unchanged checkpoint, and benchmark performance does not prove clinical substitution. The timeline and examples draw on the following sources [7, 16, 39, 49, 76, 203, 204, 207–220].

13 Multitask

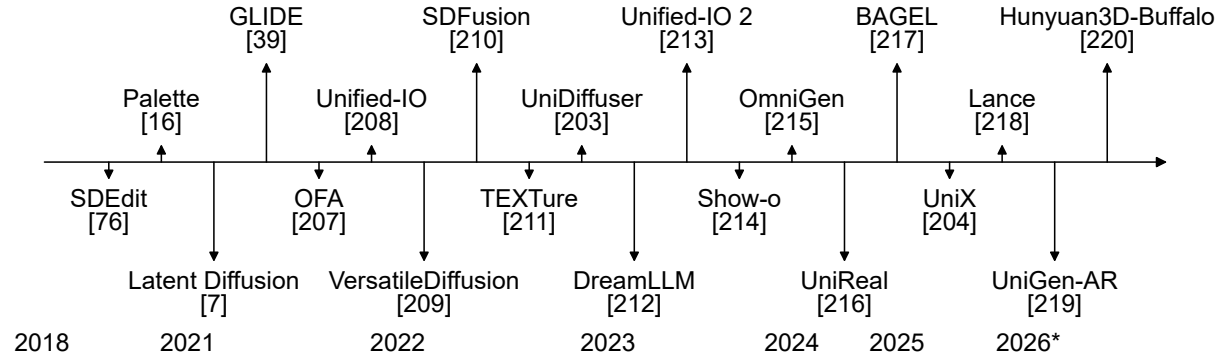


Figure 18: **13 Multitask: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.4.2 14 Multimodal

Multimodal. Multimodal generation connects semantically different representations, such as language and images. In ML, generating an image from text models a cross-modal conditional distribution; the text may remain fixed while only the image is sampled. UniDiffuser illustrates the CV distinction because the same framework supports image-to-text, text-to-image, and paired image-text generation through different sampling settings. UniX supplies a medical example with chest-radiograph report generation and text-guided radiograph synthesis. The invariant to test is agreement about the requested findings, not pixelwise identity with an individual patient. Multimodal conditioning alone is not joint sampling, and several MRI contrasts do not by themselves establish an any-to-any system. Required checks include finding-level omissions, contradictions, and unsupported details; a realistic image or fluent report is not clinical proof. The timeline and examples draw on the following sources [37, 39, 41, 203, 204, 212–214, 217, 218, 221–230].

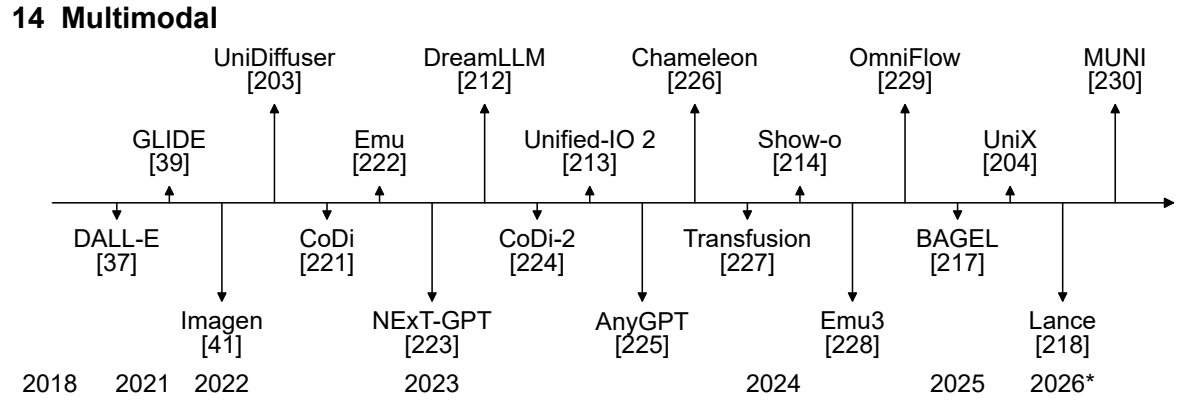


Figure 19: **14 Multimodal: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.4.3 15 Joint generation

Joint generation. Joint generation concerns dependence between outputs belonging to one sample. In ML, learning two plausible marginal distributions is insufficient: an image and its annotation must agree when sampled together. JointNet illustrates this in CV by coupling RGB and dense-modality diffusion branches to generate corresponding RGB-depth outputs. MedSegFactory provides a medical counterpart with two streams for images and segmentation masks, connected by joint cross-attention during denoising. Coupling can also use a shared latent or a dependent pipeline; simultaneous execution is not the defining requirement. Required checks assess each output and their correspondence, including boundaries, label semantics, and downstream performance on independent real cases. A jointly generated mask is not independent ground truth: both streams may reproduce the same anatomical error, and agreement alone cannot establish clinical validity. The timeline and examples draw on the following sources [203, 205, 206, 221, 229–244].

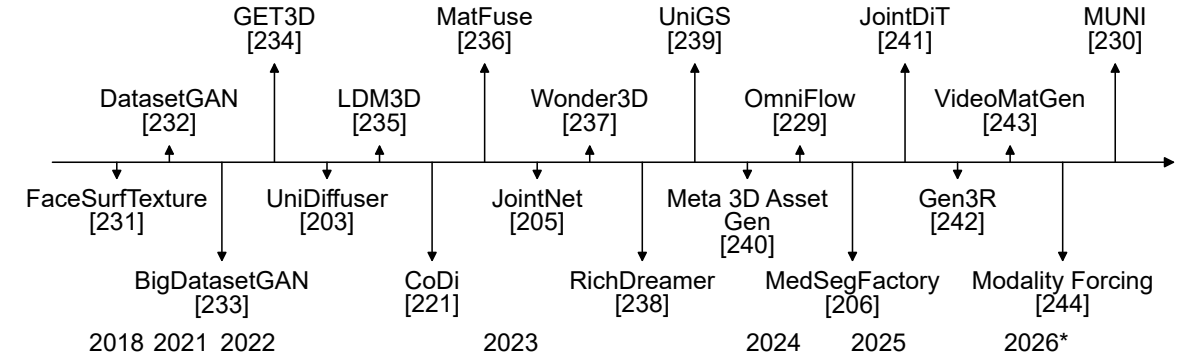


Figure 20: **15 Joint generation: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.5 Generated object

Objects specify what the output represents: an intensity array, appearance on a surface or domain, or geometry. A 3D CT volume remains an image, while a segmentation encodes anatomy. Tissue appearance is a medical texture analogue, not evidence of faithful surface texturing.

Figure 21 gives the examples for this coordinate of Figure 1.

05 Generated object

Figure 1 coordinate 5 | Subitems 16, 17, 18

Machine learning	Computer vision	Medical image analysis
16 Image	**— StyleGAN face images**	**— MAISI synthetic CT**
Noise -> sampled intensity array Represent image appearance Array meaning matters, not size Test coverage / condition fit	Latent + noise -> RGB face Vary pose and visual appearance Raster output, not explicit mesh Test diversity and distribution	Masks + noise -> 3D CT image Vary spacing and image size Target anatomy and CT intensity Test organ fit and intensity bias
17 Texture	**— Text2Tex surface texture**	**— Virtual stain analogy**
Noise / example -> texture Model appearance statistics Geometry is fixed in this example Test correlation and repetitions	Mesh + text -> texture map Paint from multiple viewpoints Keep supplied mesh geometry Test seams / view consistency	Autofluorescence -> stain image GAN changes tissue appearance Cell layout, not a mesh UV map Test matched stained structures
18 Shape	**— PointFlow point clouds**	**— CAN-FLOW cardiac anatomy**
Latent -> sampled geometry Points / mesh / occupancy / SDF Specify coordinates and topology Test validity and coverage	Latent -> new 3D shape Sample points within each shape No mesh connectivity specified Test surface coverage / diversity	Age / sex / BMI -> heart shape Sample biventricular anatomies Target metadata-related variation Test phenotypes / shape validity

Figure 21: **Generated object: ML–CV–medical correspondences.** Figure 1, items 16–18. Examples and evaluation requirements; see Section 1. Sources: [8, 9, 13, 28, 31, 245, 246].

3.5.1 16 Image

Image. Image specifies the generated object, not the learning mechanism or task. In ML, a model samples an array whose values represent intensity or color; generation can be unconditional, conditioned, or part of restoration. StyleGAN supplies a CV example by synthesizing face images with controllable appearance variation. MAISI demonstrates the medical target by using latent diffusion to synthesize three-dimensional CT, with organ segmentations available as ControlNet conditions. An intensity volume remains an image even though it occupies three-dimensional space; a segmentation volume instead encodes anatomical labels or shape. Required checks include anatomical agreement, intensity bias, diversity, and memorization rather than visual realism alone. An organ mask is a control signal, not clinical evidence, and a synthetic CT must not be treated as an acquired patient measurement. The timeline and examples draw on the following sources [7, 9, 30–34, 36–41, 44, 46, 47, 159–161, 219, 247].

16 Image

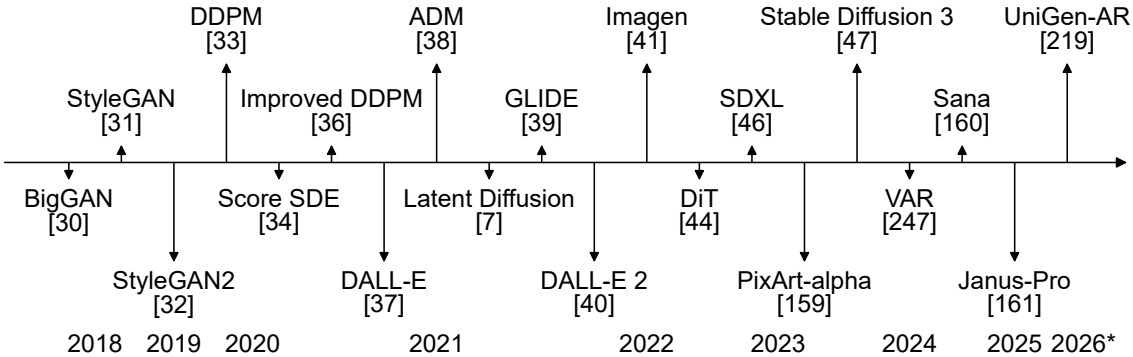


Figure 22: **16 Image: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.5.2 17 Texture

Texture. Texture names generated appearance, which may be a spatial pattern, a mesh color map, or a material field. The ML problem models spatial appearance statistics rather than geometry alone. Text2Tex is a direct CV example: a supplied mesh and text guide multiview diffusion inpainting to build its texture. Mesh geometry is the fixed support; seams and cross-view agreement need checking. The medical bridge here is explicitly an appearance analogy: virtual staining maps label-free tissue autofluorescence to a stained-looking microscopy image using a GAN. It generates image appearance, not a UV texture for a mesh. Likewise, MIVAE couples MRI appearance with cardiac shape; that does not make MRI intensity a surface material. Matching histological structures and disease-relevant details requires independent validation beyond visual realism. The timeline and examples draw on the following sources [8, 13, 211, 231, 234, 236, 243, 245, 248–261].

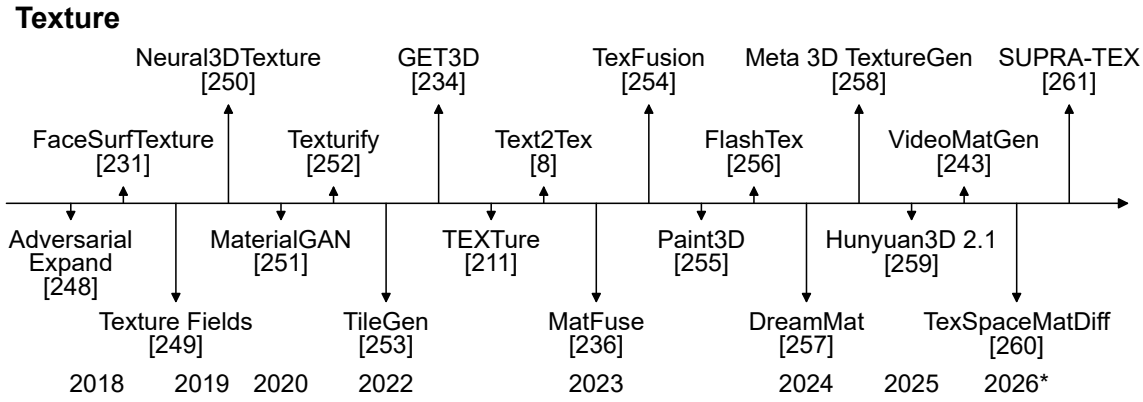


Figure 23: **17 Texture: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.5.3 18 Shape

Shape. Shape specifies a geometry-bearing output, not a particular generator. In ML, learning a distribution of coordinates, occupancies, or signed distances permits new spatial configurations to be sampled. PointFlow illustrates the CV case by modeling a distribution of shapes and a conditional distribution of points for each shape. Its point clouds do not themselves provide triangle connectivity. CAN-FLOW supplies a medical example: a conditional normalizing flow models metadata-dependent cardiac shape representations and generates biventricular anatomies. It is not a flow-matching method merely because its name contains flow. Required checks cover anatomical validity, phenotype distributions, subgroup variation, and geometric representation conventions. Evidence from healthy training anatomies does not establish diseased anatomy coverage, physiological function, or simulation readiness. Appearance and temporal motion remain separate, potentially overlapping coordinates. The timeline and examples draw on the following sources [28, 210, 220, 234, 246, 262–277].

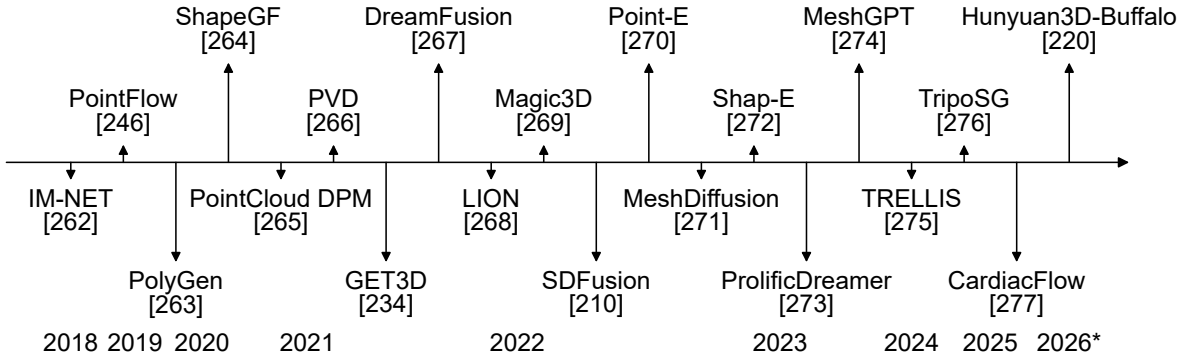


Figure 24: **18 Shape: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.6 Temporal structure

Temporal structure asks how states depend on time and actions. A time-indexed 3D object, a missing intermediate frame and a future observation under a chosen action require different references. Their smoothness can be similar while their physical or control meaning differs.

Figure 25 gives the examples for this coordinate of Figure 1.

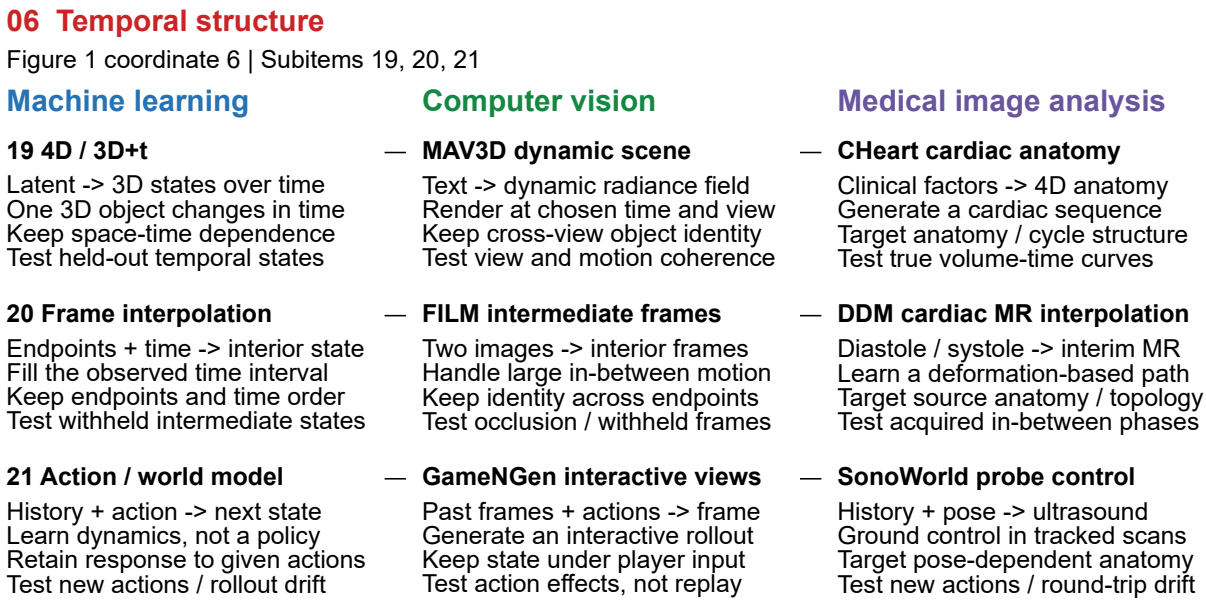


Figure 25: **Temporal structure: ML–CV–medical correspondences.** Figure 1, items 19–21. Examples and evaluation requirements; see Section 1. Sources: [24, 26, 278–282].

3.6.1 19 4D / 3D+t

4D / 3D+t. This coordinate asks whether three spatial dimensions evolve over physical time, rather than whether a sampler runs for many numerical steps. At the ML level, a model generates dependent spatial states indexed by time; independent volumes do not establish a coherent trajectory. In CV, MAV3D converts text into a dynamic 3D radiance field that can be rendered from different viewpoints. In medicine, CHeart conditions generated cardiac anatomical sequences on clinical factors. These examples illustrate an output structure, not an exclusive model family, and can also carry conditioning or mechanism labels. Validation should compare anatomy and motion against held-out reference sequences, including ventricular volume trajectories where appropriate. Smoothness or plausible population variation does not establish measured patient physiology, electrical activation, or a valid individual digital twin. The timeline and examples draw on the following sources [23, 24, 145, 278, 280, 283–297].

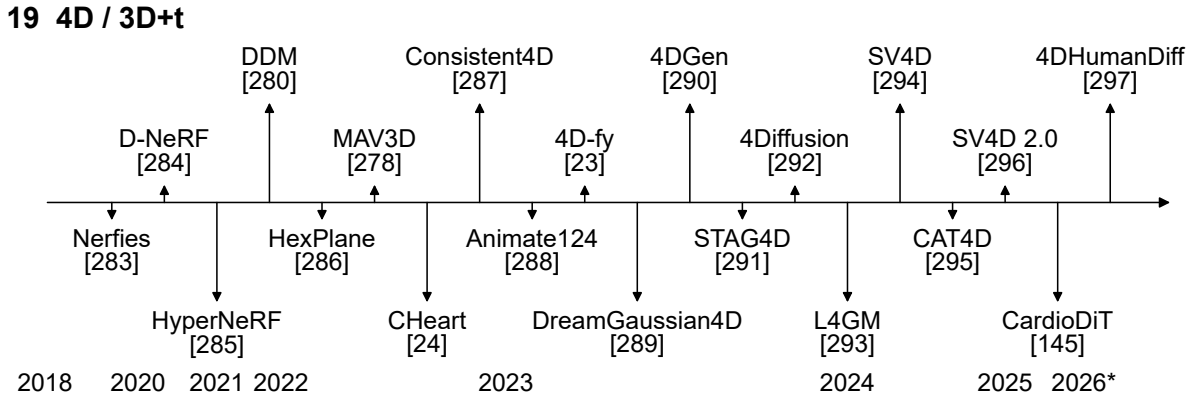


Figure 26: **19 4D / 3D+t: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.6.2 20 Frame interpolation

Frame interpolation. Frame interpolation fills times strictly between observed endpoints; extrapolation predicts beyond them. The ML problem is endpoint-conditioned synthesis, potentially with several plausible intermediate paths. FILM supplies the CV example by synthesizing intervening image frames even with large motion. The medical DDM example instead generates intermediate 3D cardiac MR volumes between diastolic and systolic observations using learned deformation. Thus the same temporal coordinate can describe ordinary image frames or spatial volumes, and it overlaps 4D generation when the latter evolve over time. Evaluation should withhold genuinely acquired intermediate phases and check anatomy, timing, and motion, rather than score endpoint reproduction alone. Interpolating a visually smooth heartbeat does not determine that patient's actual intermediate physiology or justify predicting disease progression between clinical visits. The timeline and examples draw on the following sources [279, 280, 298–315].

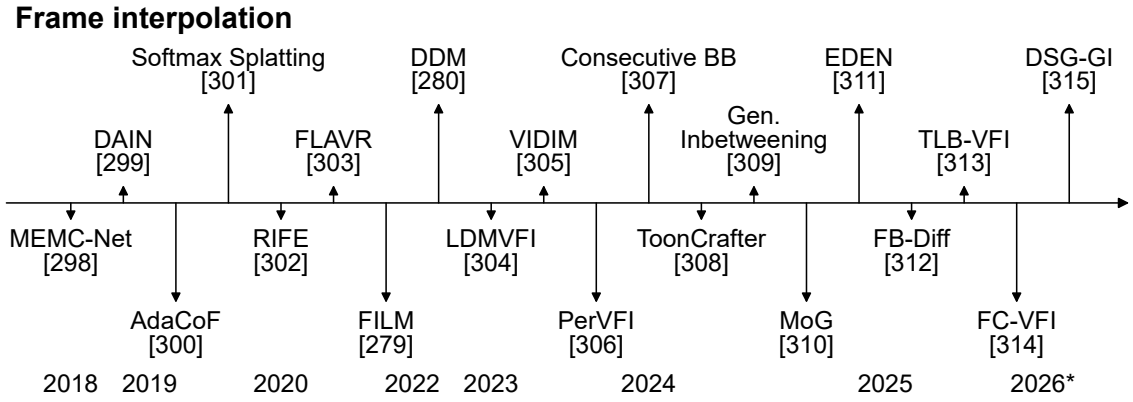


Figure 27: **20 Frame interpolation: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.6.3 21 Action / world model

Action / world model. An action-conditioned model predicts what observation or state follows a specified action and history; a policy instead chooses the action. World Models illustrates the ML formulation with a probability distribution over the next latent state conditioned on the current action and memory. GameNGen turns past frames and player actions into the next game frame. SonoWorld is a documented ultrasound preprint that grounds video generation in metric probe poses using EM-tracked freehand scans. Its pose-control evaluation is not proof that every generated response to a new action is physically valid. Validation should compare held-out action responses, opposite-direction controls, and accumulated rollout drift. Replaying recorded poses or producing convincing videos does not establish causal intervention validity, safe robotic control, or clinical benefit. The timeline and examples draw on the following sources [22, 26, 281, 282, 316–331].

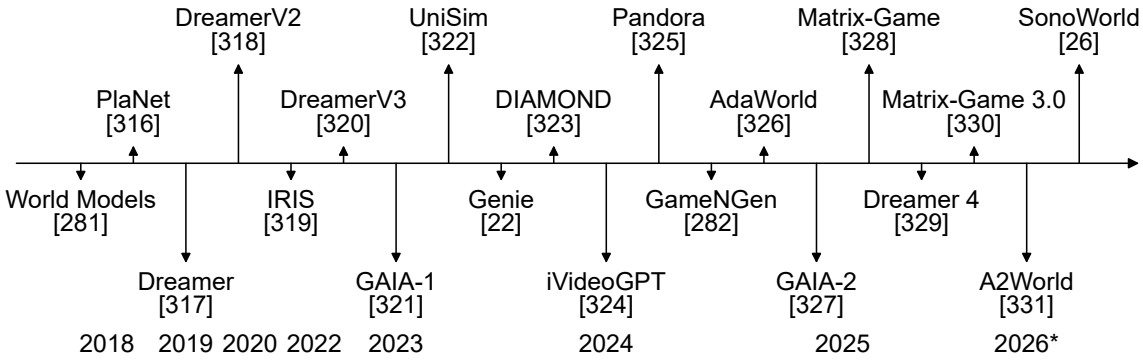


Figure 28: **21 Action / world model: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.7 Modeling mechanism

Mechanisms specify objectives and probability paths, not tasks or backbones. GAN, diffusion and flow matching are the three plotted anchors, not an exhaustive mechanism taxonomy. VAEs, autoregressive models and normalizing flows remain essential foundations [4, 33, 34, 332–335]. The following histories explain what changes mechanistically without implying that a newer family automatically preserves more information.

Figure 29 gives the examples for this coordinate of Figure 1.

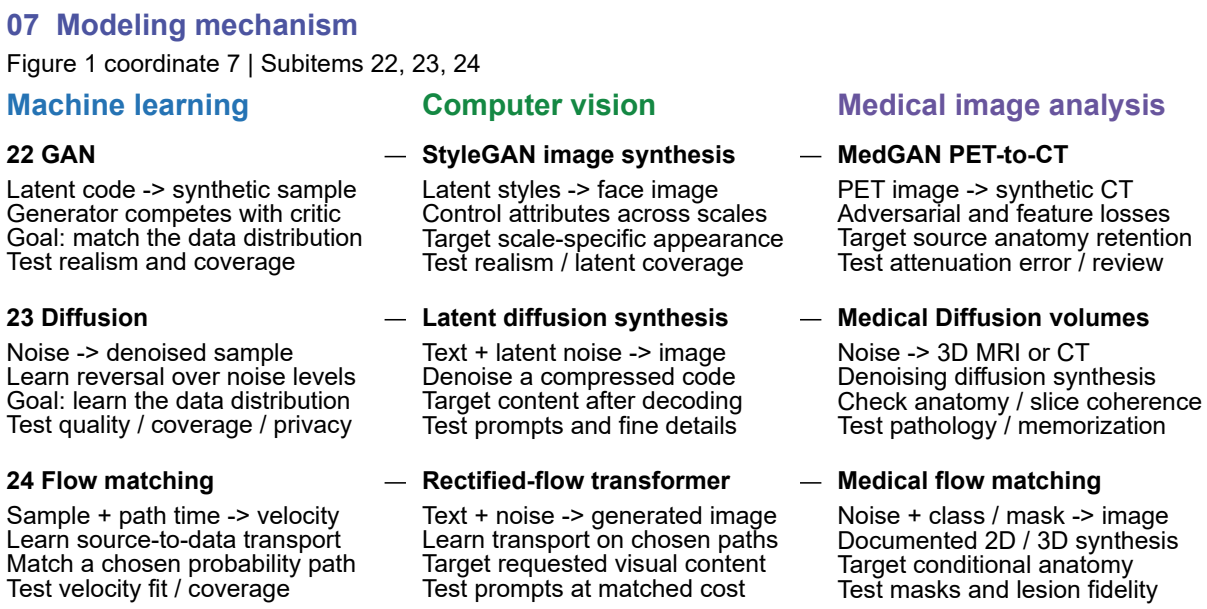


Figure 29: **Modeling mechanism: ML–CV–medical correspondences.** Figure 1, items 22–24. Examples and evaluation requirements; see Section 1. Sources: [7, 31, 33, 34, 43, 47, 53, 332, 335, 336].

3.7.1 22 GAN

GAN. A GAN is defined by adversarial learning between a generator and discriminator, not by a specific network architecture or imaging task. The ML example maps random latent codes into samples while discriminator feedback supplies a learned training signal. StyleGAN applies this principle to image synthesis with scale-specific style control. MedGAN applies adversarial, feature, and style losses to PET-to-CT translation for attenuation correction rather than direct CT diagnosis. Here the source scan constrains the desired anatomical content, but those losses do not guarantee that lesions or quantitative attenuation values survive translation. Evaluation should measure distribution coverage as well as realism, then check paired anatomy, quantitative errors, and task-specific reader judgments. A discriminator score is neither diagnostic confidence nor evidence that all clinically important modes are represented. The timeline and examples draw on the following sources [30–32, 52, 53, 58, 92, 147, 173, 193, 198, 332, 337–345].

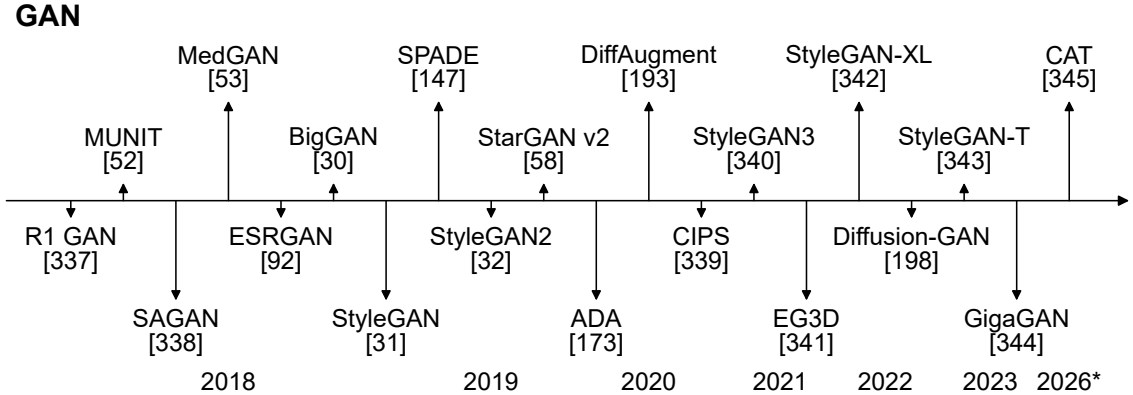


Figure 30: **22 GAN: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.7.2 23 Diffusion

Diffusion. Diffusion models learn how to reverse progressive corruption, commonly through denoising or noise-dependent score estimation. The ML example starts from noise and repeatedly applies the learned reverse process. Latent diffusion performs this process in a compressed representation and decodes it into an image, with text or other inputs providing optional conditions. Medical Diffusion documents three-dimensional MRI and CT synthesis. Its abstract reports reader assessment of appearance, anatomical correctness, and slice consistency; this is narrower than diagnostic equivalence or privacy protection. Validation should separately assess anatomical structure, clinically relevant findings, memorization, and performance on independent clinical tasks. Numerical sampling time is not patient time. A deterministic probability-flow ODE can also sample a score-trained diffusion model, so using an ODE alone does not establish flow-matching training. The timeline and examples draw on the following sources [7, 19, 33, 34, 36, 38–41, 43, 44, 46, 97, 149, 346–351].

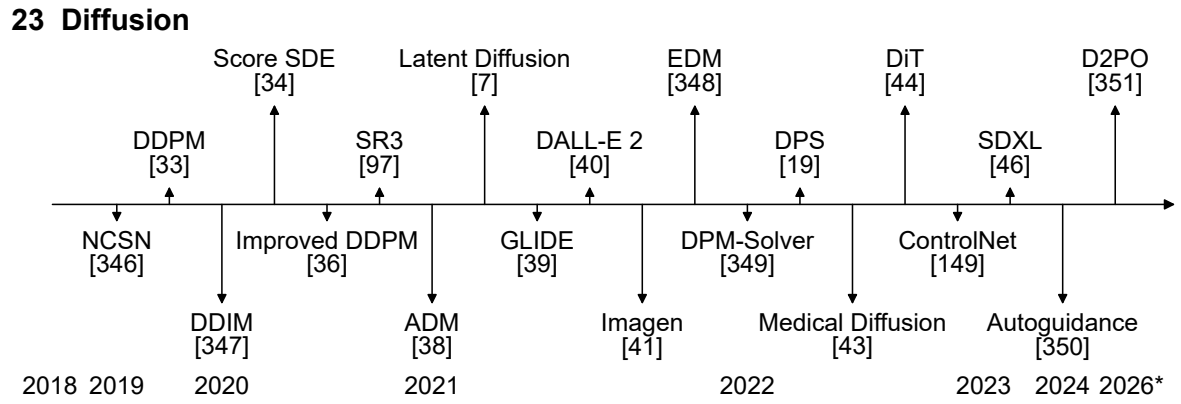


Figure 31: **23 Diffusion: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.7.3 24 Flow matching

Flow matching. Flow matching trains a velocity field against targets defined along chosen source-to-data probability paths. At the ML level, the input is an intermediate sample and path time, and the output is a transport direction used to generate new samples. Rectified-flow transformers provide the CV example through text-to-image synthesis. Flow Matching for Medical Image Synthesis documents an optimal-transport variant for 2D and 3D outputs with class or mask conditioning. This is a training coordinate, not a synonym for any invertible network, optical-flow estimator, or ODE solver. It can share probability paths and sampling machinery with diffusion. Validation should check transport accuracy, condition adherence, and anatomy at a declared sampling budget. The abstract-supported medical application does not establish clinical benefit, exact anatomical preservation, or universally faster generation. The timeline and examples draw on the following sources [47, 335, 336, 352–368].

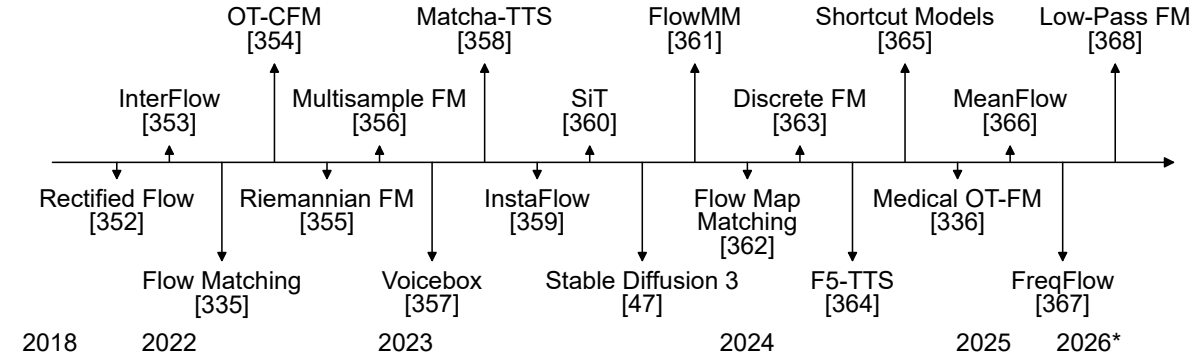


Figure 32: **24 Flow matching: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.8 Requirements and evaluation

Requirements concern what must remain correct and what resources the useful workflow consumes. Fidelity needs a reference, efficiency a budget, and data consistency a forward model with noise assumptions. These complementary tests cannot substitute for one another, and a favorable average may conceal consequential failures.

Figure 33 gives the examples for this coordinate of Figure 1.

08 Requirements and evaluation

Figure 1 coordinate 8 | Subitems 25, 26, 27

Machine learning	Computer vision	Medical image analysis
25 Fidelity	**GenEval compositional checks**	**Pathologist-guided alignment**
Real / synthetic data -> scores	Prompt + image -> alignment test	Synthetic tissue -> expert review
PRDC: fidelity versus coverage	Check count, color and position	Reward-guided diffusion tuning
Declare the reference features	Target requested object relations	Target meaningful tissue features
Test outliers / missing modes	Test attribute binding, not looks	Test independent diagnostic truth
26 Efficiency	**DPM-Solver sampling**	**MAISI-v2 volume synthesis**
Same workload -> cost / quality	Trained diffusion prior -> image	Noise + conditions -> 3D image
Time / memory / training budget	Use fewer model evaluations	Rectified flow speeds synthesis
Keep a declared quality target	Check quality at each budget	Check anatomy and condition fit
Test full runtime, same hardware	Test wall time, not step count	Test total time, failures, retries
27 Measurement consistency	**DDNM null-space restoration**	**Score-based CT / MRI**
Data + operator -> reconstruction	Linear observation -> image	Measured data -> image draws
A prior fills unobserved parts	Generate null-space content	Supply the operator at test time
Respect the stated noise model	Keep noiseless measurements	Target noise-aware data fit
Test residual / unresolved detail	Test data fit and structure	Test residuals / hidden anatomy

Figure 33: **Requirements and evaluation: ML–CV–medical correspondences.** Figure 1, items 25–27. Examples and evaluation requirements; see Section 1. Sources: [19, 20, 27, 117, 349, 369–371].

3.8.1 25 Fidelity

Fidelity. Fidelity must name its reference: a data distribution, an input condition, a perceptual target, or a clinically meaningful feature. At the ML level, the density and coverage metrics associated with PRDC distinguish sample fidelity from distribution coverage rather than collapsing them into one score. GenEval supplies the CV example by checking generated object counts, colors, and spatial relationships against text prompts; it is an evaluator, not a generator. The medical human-feedback framework uses pathologist judgments, a learned reward model, and diffusion fine-tuning to improve clinical plausibility. A convincing texture can still conceal an incorrect cell type or missing abnormality. Evaluation should therefore include independent expert and task-specific checks under patient-separated testing. Expert preference or better generic image scores alone does not establish diagnostic fidelity or prospective utility. The timeline and examples draw on the following sources [38, 369–387].

25 Fidelity

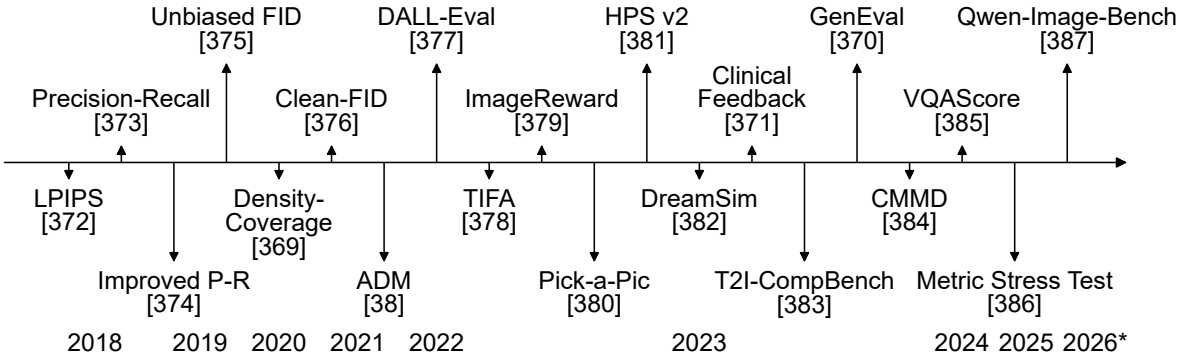


Figure 34: **25 Fidelity: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.8.2 26 Efficiency

Efficiency. Efficiency asks which resource is saved at what output quality, not simply how many sampling steps are removed. The ML comparison fixes data, hardware, and a quality target before reporting latency, memory, or training cost. DPM-Solver illustrates CV inference acceleration through a specialized solver for diffusion ODEs without additional training. MAISI-v2 documents accelerated 3D medical image synthesis using rectified flow and a region-specific contrastive loss; it is not merely DPM-Solver applied to medical scans. These mechanisms can occupy other coordinates at the same time. Evaluation should include encoding, decoding, loading, postprocessing, quality control, failures, and retries when measuring the total workflow. A fast sampling kernel or fewer network evaluations does not prove lower end-to-end cost, preserved lesion information, or diagnostic noninferiority. The timeline and examples draw on the following sources [7, 27, 36, 45, 347–349, 352, 366, 388–399].

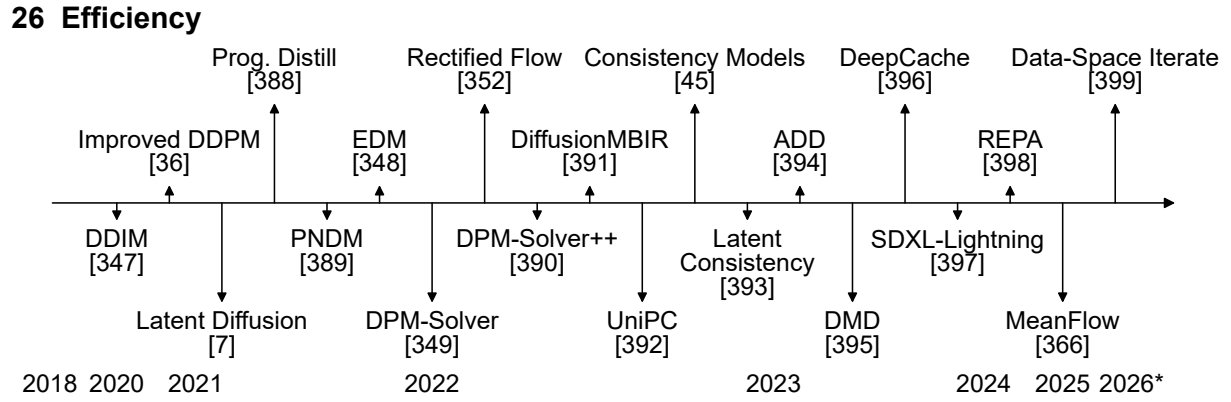


Figure 35: **26 Efficiency: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

3.8.3 27 Measurement consistency

Measurement consistency. Measurement consistency tests a reconstruction against acquired data through a specified forward operator and noise model. The ML example constrains observed components while a generative prior supplies unresolved ones. DDNM illustrates the CV case by modifying null-space content during diffusion-based linear image restoration. The medical score-based inverse framework reconstructs CT or MRI from measurements using a physical measurement model at test time. DPS additionally illustrates why noisy observations need a noise-aware objective rather than an indiscriminate exact-fit projection. Validation should examine residuals using the stated noise statistics and compare clinically relevant structures against independent references. A small residual cannot identify anatomy in an operator's null space: different images may produce identical observations. Neither measurement fit nor a plausible posterior sample alone establishes unobserved truth or calibrated uncertainty. The timeline and examples draw on the following sources [17, 19, 20, 100, 114–117, 121, 122, 124, 391, 400–407].

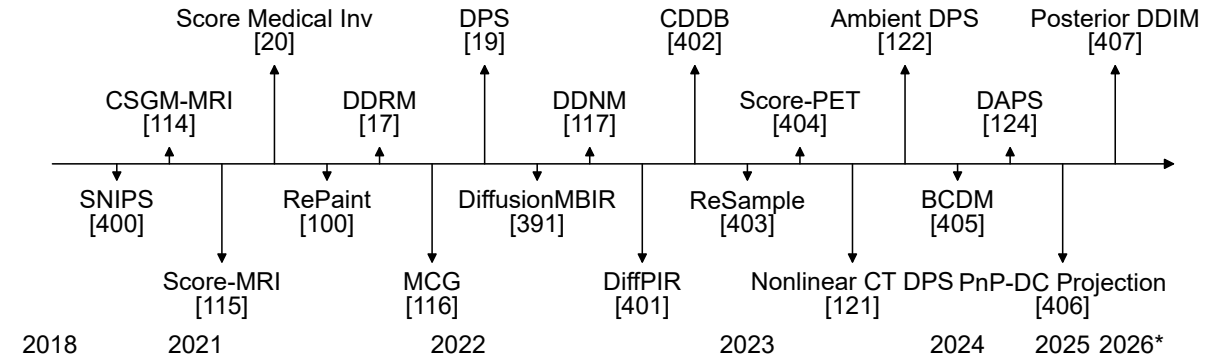


Figure 36: **27 Measurement consistency: representative publication timeline.** Twenty chronological landmarks. Dates and selection: Section 1.

4 Cross-Dimensional Trends and Evidence Synthesis

4.1 What Changes, and What the Curves Cannot Establish

Mechanisms change faster than the underlying questions. Figure 1 shows declining GAN vocabulary and rising diffusion vocabulary before the current-year endpoint, but synthesis, reconstruction and control remain distinct operations under either mechanism. In complete-frame binary mentions, GAN falls from 176/428 (41.12%) in 2018 to 226/6,219 (3.63%) in 2025; diffusion rises from 0/428 to 2,353/6,219 (37.84%). Flow matching appears in 231/6,219 records (3.71%) in 2025. These shifts describe the query's language, not a controlled comparison of accuracy, cost or clinical usefulness. A reader choosing a reconstruction problem should first identify its information and error constraints, then select a mechanism, rather than reverse that order.

2026 emphasizes action-related language, but is not a complete-year growth test. Among 7,019 YTD records, temporal generation matches 1,867 (26.60%), action conditioning 2,086 (29.72%), diffusion 1,544 (22.00%) and flow matching 401 (5.71%). Mean concept TF–IDF is 0.1803 for action conditioning versus 0.0739 in 2025, and 0.0600 for diffusion versus 0.1068. Query composition, indexing and writing choices can change these quantities. In particular, the action concept includes work outside temporal-generation matches, and the fixed 102-concept normalization couples panel magnitudes. A higher action curve does not show that models have learned useful interventions; that requires action-conditioned tests against replay and other strong controls.

Robust direction is more defensible than precise field share. Historical prevalence sensitivity places the 2018–2025 GAN change between -50.17 and -20.47 percentage points, diffusion between $+12.08$ and $+58.30$, and flow matching between $+1.51$ and $+5.71$. These are specification ranges, not confidence intervals. Direction survives these tested definitions, while magnitude does not. The 96-record relevance audit explains why interpreting a single filtered denominator as the literature would be misleading. Consequently, this survey uses counts to locate questions for evidence inspection, not to rank the scientific value of research topics.

4.2 Combinations Matter More Than Isolated Keywords

Cross-coordinate matches show which combinations deserve reading. We preselected five concept pairs spanning mechanism, condition, data regime, organization, operation and requirements, then re-evaluated the frozen rules over all 25,056 abstracts. All 918 annual concept counts reproduced exactly. Table 2 divides joint matches by anchor matches, so its denominator differs from the atlas and each row answers a specific conditional question. It is a diagnostic view of written claims, not evidence that the corresponding components were implemented together.

Table 2: **Cross-coordinate co-mentions in the fixed visual-generation frame.** Each entry is both-label matches divided by first-label matches in that year (percentage). The full yearly frames contain 6,219 records in 2025 and 7,019 in 2026*. *YTD retrieved 21 October. Labels are overlapping title/abstract matches, not verified method adoption; the years are not matched-calendar exposure.

Anchor / partner	2025	2026*
Diffusion / text-conditioned	1243/2353 (52.8%)	753/1544 (48.8%)
Temporal generation / action	139/1519 (9.2%)	369/1867 (19.8%)
Multimodal / text-conditioned	520/1089 (47.8%)	422/1040 (40.6%)
Few-shot / text-conditioned	35/77 (45.5%)	27/68 (39.7%)
Measurement inverse / data consistency	8/117 (6.8%)	10/123 (8.1%)

The action shift is not just more temporal papers. Action terms occur in 139/1,519 temporal-generation matches (9.2%) in 2025 and 369/1,867 (19.8%) in 2026 YTD. This supports inspecting the transition from producing sequences to predicting consequences of supplied actions. It does not identify which actions are measured, whether hidden state is observed, or whether a policy transfers. By contrast, text-conditioned terms occur in roughly half of diffusion matches in both snapshots (52.8% and 48.8%): diffusion and text-to-image should not be treated as synonyms. Similar care applies to multimodal and few-shot rows; their text co-mentions do not establish any-to-any generation or a comparable pretraining budget.

Sparse vocabulary is a search warning, not an empty field. Explicit data-consistency phrases co-occur in only 8/117 measurement-inverse matches in 2025 and 10/123 in 2026. This cannot support the claim that most reconstruction methods ignore measurements: likelihoods, forward operators, projections and domain-specific terms may express the same idea without these words. Likewise, small joint-generation or interpolation curves do not establish unstudied applications. A candidate gap must survive synonym expansion, direct-neighbor reading and a check of whether the proposed improvement is already expressed under another coordinate.

4.3 From Capability to Supported Use

There is no single scalar maturity of a subtask. Implementation, external image evaluation, clinical-reader agreement, workflow feasibility and patient outcomes answer different questions. Table 3 attaches these distinctions to named studies. It records what selected studies actually test, rather than assigning their evidence to all papers sharing a task label. A clinical non-diffusion MRI comparator can establish that a particular accelerated workflow is testable, but cannot validate a new diffusion sampler. Conversely, a recent generative method need not be clinically deployed to offer a useful, bounded technical result.

These selected sources span different operations and levels of validation. They are not a representative sample of all 27 subitems, and their limitations must remain attached to the named workflow rather than generalized to a whole mechanism.

Table 3: **Source-specific evidence ledger.** Scope: selected medical studies and clinical comparators, not a field-wide maturity ranking. Inspection locations and hashes are in the evidence log.

Source / use	What is evaluated	What does not follow
MAISI [9]	Conditional volumes; external autoPET image-distribution and downstream segmentation tests.	Diagnostic benefit, comprehensive application coverage or a privacy guarantee.
Medical score inversion [20]	CT/MRI inverse tasks with simulated projections or single-coil measurements.	Prospective scanner or independent reader validation.
Accelerated knee MRI [408]	Prospective 170-participant, six-reader interchangeability study.	Surgical truth, cross-vendor performance or validation of new diffusion methods.
MR-only brain RT [12]	Workflow completed in 20/21 cases with artifact-related fallback.	Universal synthetic-CT reliability or improved patient outcomes.
UV-PARS virtual histology [409]	Diagnostic concordance and blinded quality comparison by three pathologists.	Surgical-margin or patient-outcome benefit.
CAN-FLOW [28]	Metadata-conditioned static healthy anatomy; conditional anatomy evaluation.	A complete simulation workflow or general clinical validity.
Cardiac Mesh Flow [25]	Dynamic mesh benchmarks and static-heart periodicity control.	Independent local motion truth or patient-specific physiology.
VICTRE [410]	Procedural/physics virtual mammography comparison; 2,986 virtual patients.	General validity of learned cohorts, malignancy diagnosis or mortality prediction.
SonoWorld [26]	Pose-grounded ultrasound; 42 subjects, 33/9 split; action-selection/replay tests.	Independent real-robot clinical policy transfer or force-valid dynamics.

Three general priorities follow across ML and CV, before medicine adds further constraints. First, test compositional generalization: can independently useful conditions be combined without breaking an invariant? Second, compare total resources, including pretraining access, target examples and per-sample inference, rather than calling a frozen or small adapter universally efficient. Third, choose an evaluator that can disagree with the generator. These are synthesis priorities, not verified novel research gaps. JointNet already couples visual outputs, MedSegFactory generates medical image–mask pairs, and CHeart and Cardiac Mesh Flow already model dynamic anatomy [24, 25, 205, 206]. Merely joining outputs or adding time is therefore not enough.

Independent truth is the scarce resource behind several apparent gaps. An image and generated mask can share the same wrong boundary; a smooth periodic sequence can be static; a large generated dataset can still fail to represent its evaluation setting. MedSegFactory's inspected design is two-dimensional, and processed segmentation or registration references do not become independent motion truth because they are dense [25, 206, 245]. The useful next question is which wrong output the current evaluator would accept. Medical applications sharpen that question by linking the error to a measurement, diagnosis, treatment quantity or validated trial endpoint.

5 Medical Needs and Validation

A medical task starts from the decision that an error could change. The relevant chain is target population, acquisition, generated output, interpretation, action and endpoint. A plausible image is useful only if the required link is preserved. Figure 37 illustrates three selected medical workflows, not the entire taxonomy and not a generic maturity ladder. It complements the correspondence maps: those maps explain meanings, whereas this figure locates validation obligations after a medical use has been chosen.

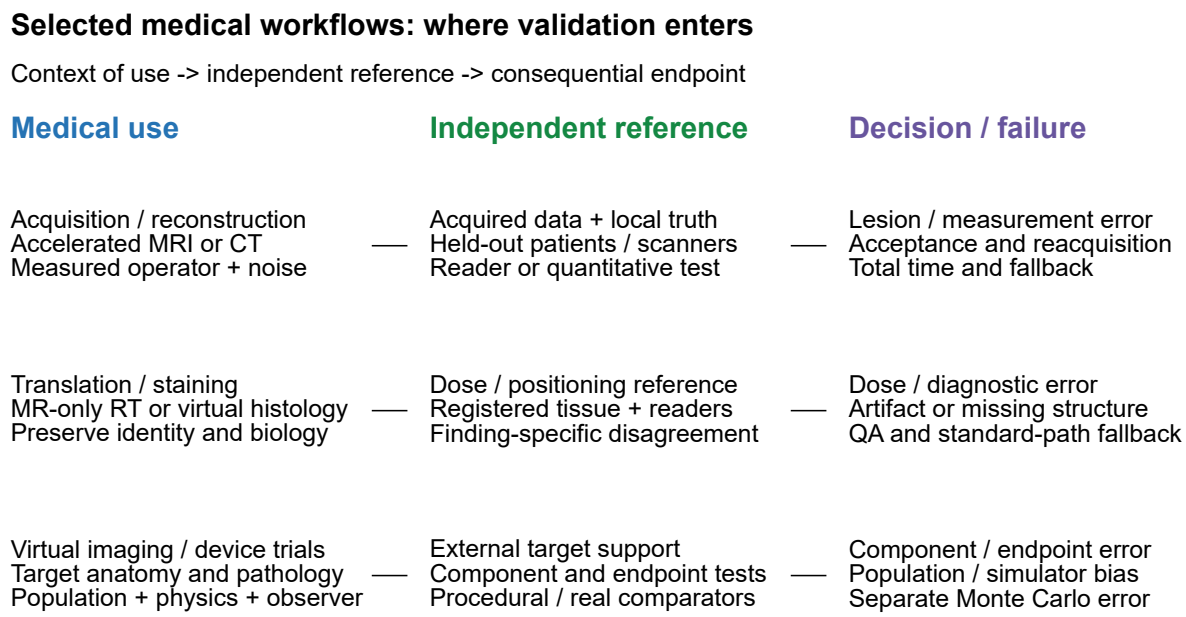


Figure 37: **Selected medical validation paths.** Scope: three use families, not all generative AI or one temporal subtask. Independent references connect generated outputs to a consequential endpoint. Sources: [12, 20, 28, 408–412].

5.1 Acquisition and Reconstruction: Preserve Decision-Relevant Information

Faster acquisition is valuable only if important errors remain acceptable. Reconstruction studies need raw-data splits, realistic operator shifts, strong deterministic and generative comparators, and prespecified lesion or quantitative endpoints. The prospective knee MRI study used 170 participants and six readers to test interchangeability, but lacked surgical truth and used a single vendor [408]. It is a non-diffusion comparator, demonstrating an evaluation design rather than validating newer priors. Data-consistent samples can still differ in poorly observed structures; hallucination work already makes this problem explicit [18, 89]. A useful uncertainty output should identify cases requiring another measurement or fallback, with the acquisition and review burden included.

Translation becomes consequential when a downstream quantity is used. For MR-only radiotherapy, synthetic CT supports dose calculation and positioning, not merely CT-like appearance. An inspected brain RT workflow completed 20/21 cases and exposed an artifact-related fallback associated with hemostatic material [12]. Its findings attach to that workflow; separate atlas-based synthetic-CT quality assurance is not a universal learned-generator certificate [411]. SynthRAD separates image, geometry and dose criteria [50]. This makes a practical research question precise: does a method preserve dose-relevant quantities under the scanner, pathology and implant conditions that cause failures, and can it recognize when it does not?

5.2 Diagnostic Translation: Preserve the Structure Behind the Label

Virtual staining must preserve diagnostic evidence, not just a staining style. Registration, tissue preparation and reference quality affect paired comparisons. UV-PARS reports diagnostic concordance and a blinded three-pathologist image-quality comparison, but not improved surgical margins or patient outcomes [409]. Recent context-of-use discussion is relevant to validation design, yet the neighbor inspected only at abstract level cannot carry a detailed empirical

claim [412]. A useful study separates finding-specific omissions, introduced structures and unreadable cases, tests tissue and site shifts, and retains pathologist review. Averaging these errors into an attractive perceptual score would erase the reason to perform the translation.

5.3 Virtual Trials: Validate the Population and the Simulator

More virtual patients reduce sampling error only within the assumed model. VICTRE compared mammography systems using procedural breast generation, imaging physics and observers across 2,986 virtual patients, with a human-trial comparison [410]. It was not a learned-generative-AI invention and its demonstrated detection task does not validate malignancy diagnosis or mortality. Limited lesions, motion, views and target populations constrain extrapolation. CAN-FLOW illustrates conditional anatomical modeling, but its inspected static data do not establish the validity of a complete trial workflow [28]. The study's conclusions remain bounded by the assumptions and validation of its components. Increasing cohort size cannot eliminate bias in anatomy, physics, observer or sampling assumptions.

Patient-specific simulation needs more than a patient-shaped object. Virtual-heart ablation work includes retrospective and small prospective evidence, but used averaged electrophysiological parameters in parts of the workflow [413]. Dynamic geometry alone does not identify individual material, conduction or treatment-response parameters. Likewise, pose-conditioned ultrasound generation does not determine contact force, tissue deformation or scanner-setting effects [26, 127]. Appropriate intermediate targets are independently checked local motion, measured action–observation relations and validated endpoint changes. The stronger claim of treatment or policy validity should wait for the additional measurements that make it identifiable.

Validation guidance must match the model's role. FDA credibility guidance concerns computational modeling and simulation within its stated physics/mechanistic scope; it is not blanket approval guidance for all GenAI [414]. DECIDE-AI addresses early live clinical decision-support evaluation, not generic offline synthesis [415]. Both motivate explicit context of use, risk and workflow reporting, but checklist compliance does not establish model quality. The study should name which decision is being supported, what error is tolerable, which independent reference can detect it and what happens when the system declines or fails.

6 Research Considerations and Limitations

6.1 Interpret Evidence in Its Context of Use

A useful comparison starts with a clearly specified task. Similar model names can conceal different inputs, outputs, data access and evaluation conditions. Figure 38 provides a general reading checklist for interpreting such differences. An implementation, a benchmark result and a validated application are distinct types of evidence. The discussion below concerns how to read and compare existing work; it does not prescribe particular application directions or rank the 27 subitems.

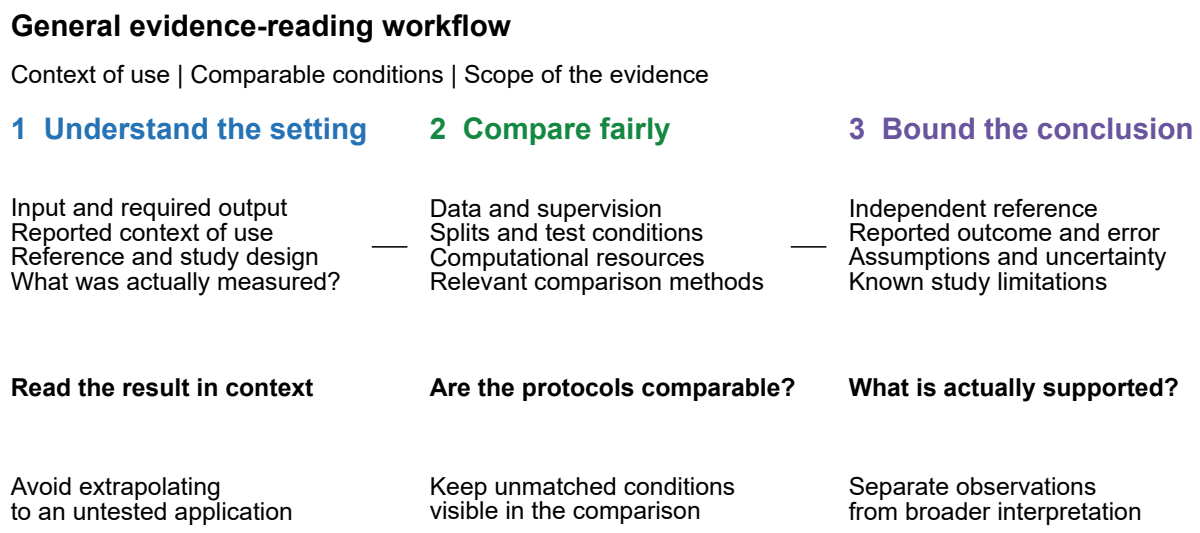


Figure 38: **General evidence-reading workflow.** Scope: interpretation of published evidence across tasks. Context, comparison and claim boundaries complement the checklist in Table 4; the diagram does not prescribe an application-specific research agenda.

Evidence should be interpreted at the level at which it was collected. A method-level result supports a statement about the tested capability, while a use-specific study supports a statement about its reported setting. Neither automatically establishes effectiveness in another population, acquisition system or workflow. Useful synthesis therefore keeps the study context, comparator and reference visible instead of compressing them into a universal ranking. Differences in available validation resources are part of the comparison, not merely reporting details.

6.2 Four General Evidence Considerations

Context of use defines what a result means. Reconstruction, hallucination assessment, diagnostic translation and measurement selection examine different questions [18, 19, 109, 412]. Recent conditional-diffusion and evaluation studies likewise report results under their own measurement, data and benchmark assumptions [416–418]. These sources are informative without being interchangeable. A survey comparison should identify the input, required output, reference and reported endpoint before discussing which method performs better. Visual similarity and task-specific performance remain separate observations.

Comparability requires the assumptions behind each result to remain visible. Joint-output systems, medical generation frameworks and dynamic representations use different training supervision and output structures [24, 25, 49, 205, 206, 245]. Their reported results should be read with the corresponding data splits, annotations and computational conditions. Shared labels can explain agreement without supplying an independent reference. A stronger conclusion needs evidence that addresses the intended claim rather than a change in architecture or representation alone. Where protocols differ, a qualitative comparison is more defensible than a single numerical ordering.

Reported performance does not remove uncertainty in the study setting. Published anatomical modeling, cohort construction and virtual-imaging studies illustrate several ways to combine data-driven and conventional components [28, 410, 419, 420]. Credibility frameworks emphasize that conclusions depend on the assumptions and validation of those components [421]. The general lesson is to distinguish what was measured, what was modeled

and what was assumed. Increasing the number of generated outputs does not by itself strengthen the validity of an application claim. Results should remain attached to the conditions under which they were obtained.

Claims should be no broader than the observations supporting them. Existing medical observation models provide concrete examples of generation conditioned on recorded inputs [26, 127]. Their scope should be described through the data and tests actually reported. A convincing output is not automatically evidence of a complete workflow benefit, and performance under one protocol does not certify another. Clear reporting separates demonstrated behavior, unresolved uncertainty and interpretations requiring additional evidence. This distinction is useful across the taxonomy rather than a recommendation for any particular project.

Table 4 summarizes four general questions for interpreting the reviewed evidence. These questions apply across tasks and do not constitute an application-specific research agenda.

Table 4: **General evidence-reading checklist.** Scope: interpretation and comparison of existing studies across the taxonomy. The rows are reporting considerations, not proposed research projects.

Aspect	Reading question	Evidence to retain	Interpretation boundary
Context	What task and setting were tested?	Inputs, outputs and context of use.	Do not infer another use from appearance alone.
Comparison	Are the conditions comparable?	Data splits, supervision, resources and baselines.	Do not rank unmatched protocols as one experiment.
Evidence	What was measured independently?	Reference origin, endpoints and uncertainty.	Agreement alone does not establish correctness.
Scope	How far does the claim extend?	Tested conditions and explicit limitations.	Do not extrapolate beyond the reported evidence.

6.3 Limitations of the Evidence Base

The survey cannot certify an empty field, a best topic or clinical readiness. The query misses differently named work and includes background mentions; the historical audit and sensitivity tests do not validate all labels or the 2026 extension. Purposive selection favors accessible and salient work. Single-AI extraction lacks independent expert adjudication, and abstract/indexed access is weaker than locally inspected passages. Timeline landmarks are not a citation ranking. No field-wide absence, global first, regulatory assessment or acceptance-probability claim follows here. New publications can also change the interpretation of a rapidly developing topic.

Transparent boundaries make the review useful beyond a list of models. Readers can use the taxonomy to locate related formulations and the evidence tables to understand what a published result supports. The essential questions are the intended task, available input, required invariant, reference, comparator and reported outcome. When one of these is unspecified, uncertainty should remain explicit rather than being replaced by a confident ranking. The resulting framework supports careful reading and evaluation without turning descriptive trends into a prescriptive research roadmap.

7 Conclusion

The eight coordinates and 27 histories separate what generation does from its conditions, resources, representations and mechanism. This makes both reuse and disagreement easier to understand: changing a model family need not change the task, while the same operation can require different references in computer vision and medical image analysis. The descriptive corpus records changes in research vocabulary without equating them with adoption or application readiness. Across the reviewed studies, useful interpretation depends on the task, comparison conditions and scope of the reported evidence. The framework brings these elements together as a compact guide to reading, comparing and evaluating generative AI beyond visual realism.

Transparency and Reproducibility

This is a Codex-assisted research draft, not an independently adjudicated clinical review. Retrieval, passage extraction, coding, synthesis and writing used the current Codex system; expert verification is required before submission or clinical interpretation. No new patient experiment, training, ethics approval or prospective registration is claimed. The archive includes source identities, inspection locations, coding rules, analysis tables, editable figure sources and compilation instructions. Third-party full-text PDFs are not redistributed. Journal rankings and popularity are not evidence grades. Limitations in Section 6.3 apply throughout.

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